# Open-FDD Documentation

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# 1 Home

# 2 Open-FDD

Open-FDD is an open-source **building edge platform** for BACnet integration, feather historian storage, Arrow-native fault detection, and operator dashboards — designed for **trusted LAN / OT edge** deployment behind Caddy, not direct public internet exposure.

## 2.1 Who it is for

| Role | What you get |
|---|---|
| **IT / controls engineer** | Pull three GHCR images, log in, poll BACnet, view faults |
| **Commissioning integrator** | Discover points, bind BRICK model, pin FDD rules in Rule Lab |

| Role | What you get |
|---|---|
| **Developer / maintainer** | Extend the Operator Bridge, publish images, run CI and LAN security scans |

## 2.2 Start here

| Path | Go to |
|---|---|
| **Try it** | Quick Start — Run with Docker |
| **Operate it** | Updating the stack · Operations |
| **Develop it** | Developer Guide · Architecture |

## 2.3 v3 edge stack

Three primary containers on a Linux edge host:

| Container | Role |
|---|---|
| **openfdd-bridge** | Operator Bridge — API, dashboard, feather **ingest** |
| **openfdd-commission** | BACnet discover/ read/write + **poll loop** |
| **openfdd-mcp-rag** | Doc-search sidecar for agent tools |

Host **Caddy** on `:80` or `:443` is the normal LAN front door. BACnet UDP `:47808` is bound by **commission** only — the bridge ingests `samples.csv` into the feather historian.

Details: Containers · Deployment modes

> **LAN / OT edge only.** Do not expose the Operator Bridge to the public internet without TLS, strong auth, and site review. BACnet writes are disabled by default — see BACnet write guard.

## 2.4 Distribution

| Channel | Use when |
|---|---|
| **GHCR** `ghcr.io/bbartling/openfdd-*` | Production or trial edge deploy |
| **This repository** | Custom builds, commissioning, Rule Lab development |

| Channel | Use when |
|---|---|
| **PyPI** `open-fdd` | Arrow runtime lint/test without full UI |

## 2.5 License

MIT — see repository `LICENSE`.

# 3 Run with Docker images

# 4 Run with Docker images

Deploy Open-FDD on a Linux edge host using **three published GHCR images**. No git clone required.

## 4.1 Images

| GHCR image | Service | Role |
|---|---|---|
| `ghcr.io/bbartling/openfdd-bridge` | `bridge` | Operator API, dashboard, historian ingest |
| `ghcr.io/bbartling/openfdd-commission` | `commission` | BACnet discover/ read/write **and poll loop** |
| `ghcr.io/bbartling/openfdd-mcp-rag` | `mcp-rag` | Doc-search sidecar |

Edge hosts pull **`latest`** from GHCR (three images: bridge, commission, mcp-rag). The retired **`openfdd-bacnet-poll`** image is no longer published.

## 4.2 1. Install and validate Docker

```
sudo systemctl enable --now docker
sudo usermod -aG docker "$USER"
newgrp docker   # or log out/in
```

```
docker --version
docker compose version
docker ps
docker run --rm hello-world
```

Expect: `active` / `enabled` for docker, `docker` in `groups`, hello-world succeeds.

## 4.3 2. Bootstrap the site (one script)

Downloads `docker-compose.yml`, creates `workspace/`, generates `auth.env.local`, and sets `BACNET_BIND` from the host LAN NIC (e.g. `ens192` → `10.200.200.185/24:47808`):

```
curl -fsSL -o /tmp/openfdd_edge_bootstrap.sh \
  https://github.com/bbartling/open-fdd/raw/refs/heads/master/
scripts/openfdd_edge_bootstrap.sh
bash /tmp/openfdd_edge_bootstrap.sh
```

Bootstrap + pull + start in one go:

```
bash /tmp/openfdd_edge_bootstrap.sh --start
```

**Auth** — bootstrap writes `~/open-fdd/workspace/auth.env.local` (gitignored). This file holds the API signing secret and one username/password per role. Use it for your first dashboard login:

```
cat ~/open-fdd/workspace/auth.env.local
```

| Variable | Role |
|---|---|
| `OFDD_AUTH_SECRET` | Signs session tokens (do not share) |
| `OFDD_OPERATOR_*` | Read-only operator |
| `OFDD_INTEGRATOR_*` | **Default dashboard login** — commissioning + BACnet |
| `OFDD_AGENT_*` | Agent / automation |

Example shape (passwords are **random on first bootstrap only** — existing file is kept on re-runs):

```
OFDD_AUTH_SECRET=<random>
OFDD_OPERATOR_USER=operator
OFDD_OPERATOR_PASSWORD=<random>
OFDD_INTEGRATOR_USER=integrator
OFDD_INTEGRATOR_PASSWORD=<random>
```

```
OFDD_AGENT_USER=agent
OFDD_AGENT_PASSWORD=<random>
```

Sign in as **integrator** with the password from that file → First login and health check.

| Action | Touches `auth.env.local`? |
|---|---|
| First `bash … bootstrap.sh --start` | Creates file with random passwords (if missing) |
| `bash … bootstrap.sh --start` again | **Keeps** your file — does not overwrite |
| `bash … bootstrap.sh --restart` | **Keeps** your file — only restarts bridge to reload it |
| `nano auth.env.local` + `--restart` | Uses **your** edited passwords |
| `--force-auth` | **Regenerates** random passwords (opt-in only) |

The **bridge** container reads `auth.env.local` at startup only. After you change passwords (manual edit or `--force-auth`), recreate bridge (env files are not reloaded by `restart` alone):

```
cd ~/open-fdd && docker compose up -d --force-recreate bridge
```

Regenerate passwords and reload (stack already running):

```
bash /tmp/openfdd_edge_bootstrap.sh --force-auth --restart --show-secrets
```

Manual edit then reload:

```
nano ~/open-fdd/workspace/auth.env.local
bash /tmp/openfdd_edge_bootstrap.sh --restart
```

Options:

| Flag | Purpose |
|---|---|
| `--start` | `docker compose pull && up -d` after layout; curls `http://127.0.0.1:8765/health` |
| `--image-tag TAG` | default `latest` |
| `--repo-ref BRANCH` | branch for `compose.edge.yml` if not on `master` yet |
| `--force-auth` | regenerate `auth.env.local` (restarts bridge if stack is up) |
| `--restart` | |

| Flag | Purpose |
|---|---|
| | recreate bridge to reload `auth.env.local`; curls `/health` after |
| `--show-secrets` | print passwords at end (lab only) |

From a repo checkout: `./scripts/openfdd_edge_bootstrap.sh --start`

The script prints **BACnet NIC**, **BACNET_BIND**, and file paths — **validate** `commission.env` before polling OT devices:

```
nano ~/open-fdd/workspace/bacnet/commissioning/commission.env
```

> If you used `--start`, the stack is already up. Next step → First login and health check.

### 4.3.1 Manual start (optional)

Use this only if you ran bootstrap **without** `--start` and want to bring the stack up yourself:

```
cd ~/open-fdd
docker compose pull
docker compose up -d
docker compose ps
curl -sf http://127.0.0.1:8765/health && echo
```

Expected: **bridge**, **commission**, **mcp-rag** — all `Up`. Then → First login and health check.

## 4.4 Long-term operation

### 4.4.1 Survive power cycles

All services use `restart: unless-stopped`. After reboot:

```
cd ~/open-fdd && docker compose ps
curl -sf http://127.0.0.1:8765/health
```

### 4.4.2 Start / stop / restart

```
cd ~/open-fdd
docker compose up -d          # idempotent
docker compose stop
docker compose restart commission
docker compose logs -f --tail 100 commission
```

> Never run `docker compose down -v` or delete `workspace/` on a live site.

### 4.4.3 LAN access

Put **Caddy** (or nginx) on port 80 → `127.0.0.1:8765`, or expose 8765 through the firewall.

## 4.5 Deployment choices

| Mode | Compose services | When to use |
|---|---|---|
| **Standard OT / BACnet** | `bridge` + `commission` + `mcp-rag` | Real buildings — commission polls BACnet into `samples.csv`; bridge ingests to feather |
| **Non-BACnet / demo** | `bridge` + `mcp-rag` (omit `commission`) | JSON API, Modbus-only, or lab without field BACnet — see Driver framework |
| **TLS on OT LAN** | Same stack + **host Caddy** `OFDD_CADDY_MODE=tls` | Encrypt browser↔edge; trust self-signed CA on operator PCs — not for public internet |

Do **not** run the retired `openfdd-bacnet-poll` image or legacy `openfdd-bacnet-poll` systemd unit alongside Docker **commission** — that double-polls BACnet.

## 4.6 Next steps

→ First login and health check
→ Updating the stack — `./scripts/openfdd_site_update.sh`

# 5 First login and health check

# 6 First login and health check

Quick checks after `docker compose up -d` on the edge host. Services use `restart: unless-stopped` — after a reboot, run `docker compose ps` once Docker is up (see Run with Docker).

## 6.1 Public health

```
curl -s http://127.0.0.1:8765/health
```

Through Caddy on port 80:

```
curl -s http://127.0.0.1/health
```

Expect `"ok": true` and `"auth_required": true` when auth is configured.

## 6.2 Login

1. Open the dashboard (`http://<host-lan-ip>/` or `:8765`).
2. Sign in with credentials from `workspace/auth.env.local`.

```
curl -s -X POST http://127.0.0.1:8765/api/auth/login \
  -H 'Content-Type: application/json' \
  -d '{"username":"integrator","password":"<from-auth-env>"}'
```

## 6.3 Smoke checklist

| Check | Pass |
|---|---|
| `docker compose ps` | bridge, commission, mcp-rag **Up** |
| `GET /health` | `200`, `ok: true` |
| Login | Token returned |
| BACnet poll | `samples.csv` growing (`tail -1 workspace/bacnet/polls/samples.csv`) |
| Historian | Files under `workspace/data/feather_store/` |

## 6.4 Stack detail (authenticated)

Integrator token required:

```
TOKEN=$(curl -sf -X POST http://127.0.0.1:8765/api/auth/login \
  -H 'Content-Type: application/json' \
  -d '{"username":"integrator","password":"..."}' | jq -r .token)
curl -sf http://127.0.0.1:8765/health/stack -H "Authorization:
Bearer $TOKEN" | jq .
```

Look for `bacnet_poll` green/yellow in the services list.

## 6.5 Troubleshooting

| Symptom | Fix |
|---|---|
| 401 everywhere | Configure `workspace/auth.env.local` |
| Empty BACnet discover | Fix `BACNET_BIND` in `commission.env` — network setup |
| No poll rows | Commission container running? `points.csv` has enabled rows? |
| LAN browser timeout | Open firewall port 80 or 8765 |

Advanced probes (model graph, Rule Lab, compose profiles): Developer health check.

# 7 Quick Start

# 8 Quick Start

Run Open-FDD on a Linux edge host using **three GHCR Docker images**. No git clone on the host.

| Step | Page |
|---|---|
| 1 | Run with Docker images — bootstrap script + `docker compose up` |
| 2 | First login and health check |
| 3 | Updating the stack — backup + pull |

**One-liner bootstrap** (edge host):

```
        curl -fsSL -o /tmp/openfdd_edge_bootstrap.sh \
          https://github.com/bbartling/open-fdd/raw/refs/heads/master/
scripts/openfdd_edge_bootstrap.sh
        bash /tmp/openfdd_edge_bootstrap.sh --start
```

**Prerequisites:** Linux, Docker enabled on boot, BACnet LAN if polling OT equipment.

> **BACnet writes are disabled by default.** See Write safety.

**Stack:** `bridge` + `commission` (BACnet + poll) + `mcp-rag`. Details: Containers.

# 9 Updating the stack

# 10 Updating the stack

Upgrade GHCR images on the edge host: **backup `workspace/`, pull new tags, recreate containers**. No git pull on the host.

## 10.1 Before you start

1. SSH to the edge host (`cd ~/open-fdd`).
2. Pick a tag from GitHub Packages — default is **`latest`**.
3. Short maintenance window (containers restart briefly; `restart: unless-stopped` brings them back after reboot).

## 10.2 Fetch helper scripts (no git clone)

If you bootstrapped with `openfdd_edge_bootstrap.sh`, scripts are already under `~/open-fdd/scripts/`. Otherwise:

```
mkdir -p ~/open-fdd/scripts
curl -fsSL -o ~/open-fdd/scripts/openfdd_site_backup.sh \
  https://github.com/bbartling/open-fdd/raw/refs/heads/master/scripts/openfdd_site_backup.sh
curl -fsSL -o ~/open-fdd/scripts/openfdd_site_update.sh \
  https://github.com/bbartling/open-fdd/raw/refs/heads/master/scripts/openfdd_site_update.sh
chmod +x ~/open-fdd/scripts/openfdd_site_*.sh
```

## 10.3 One-command upgrade (recommended)

```
cd ~/open-fdd
./scripts/openfdd_site_backup.sh
./scripts/openfdd_site_update.sh
```

Pulls **`:latest`** by default, verifies all three images exist on GHCR, recreates containers, and hits `/health`.

## 10.4 Step by step

### 10.4.1 1. Backup site state

```
cd ~/open-fdd
./scripts/openfdd_site_backup.sh
```

Archives `workspace/` (feather, BACnet CSVs, model, auth) under `~/openfdd-backups/<timestamp>/`.

Custom backup location:

```
BACKUP_ROOT=~/openfdd-backups/manual ./scripts/openfdd_site_backup.sh
```

### 10.4.2 2. Verify images (optional)

```
export NEW_TAG="${NEW_TAG:-latest}"

docker manifest inspect ghcr.io/bbartling/openfdd-bridge:${NEW_TAG} >/dev/null &&
docker manifest inspect ghcr.io/bbartling/openfdd-commission:${NEW_TAG} >/dev/null &&
docker manifest inspect ghcr.io/bbartling/openfdd-mcp-rag:${NEW_TAG} >/dev/null &&
echo "All images exist for ${NEW_TAG}"
```

### 10.4.3 3. Pull and recreate

```
cd ~/open-fdd
./scripts/openfdd_site_update.sh
```

**Manual equivalent** (same as the script):

```
cd ~/open-fdd
docker compose pull
docker compose up -d --force-recreate
docker compose ps
curl -sf http://127.0.0.1:8765/health && echo
```

### 10.4.4 4. Verify

```
docker compose ps
curl -sf http://127.0.0.1:8765/health | jq .
tail -1 workspace/bacnet/polls/samples.csv
ls -lah workspace/data/feather_store/
```

→ Health check

## 10.5 Dashboard UI updates

If the release changed the React UI, copy a new `workspace/api/static/app/` build onto the host before or after the image pull. Image pull alone may not change the browser bundle hash.

## 10.6 Rollback

Restore `docker-compose.yml` from backup and pull again, or re-run bootstrap from a known-good `master` commit. GHCR keeps only the **three newest** versions per image; early sites should rely on **`latest`** plus `workspace/` backups.

`workspace/` is untouched by image-only upgrades.

## 10.7 Safe cleanup

```
docker image prune -f
```

> Never run `docker compose down -v`, `docker volume prune`, or delete `workspace/` on a live site.

## 10.8 Next steps

→ Run with Docker images — bootstrap a new host
→ Health check

# 11 Local development

# 12 Local development

## 12.1 Clone and auth

```
git clone https://github.com/bbartling/open-fdd.git && cd open-fdd
cp workspace/auth.env.example workspace/auth.env.local
# Edit integrator/operator passwords and OFDD_AUTH_SECRET (32+ chars)
```

## 12.2 Start stack

```
./scripts/build_and_test.sh          # dashboard build + pytest
./scripts/docker_build.sh
./scripts/openfdd_stack.sh up
./scripts/stack_health_check.sh
```

| URL | Service |
|---|---|
| `http://127.0.0.1:8765` | Bridge (direct) |
| `http://127.0.0.1:8765/docs` | OpenAPI |

## 12.3 BACnet bench overlay (optional)

```
docker compose -f docker/compose.dev.yml -f docker/
compose.bench.yml up -d
# commission.env — set BACNET_BIND to your OT interface
```

## 12.4 Production-like local stack (Caddy, LAN scans)

Same shape as an Acme edge deploy: **prod React UI**, **Caddy on :80**, bridge on loopback **8765**, auth enabled. Use for OWASP ZAP or nmap from another machine on the LAN.

```
./scripts/pentest_production_stack.sh start    # build prod UI +
Caddy + auth.pentest.local
./scripts/pentest_production_stack.sh status   # bridge,
commission, caddy, mcp-rag
./scripts/pentest_production_stack.sh verify   # LAN IP, /health,
ZAP target URL
./scripts/pentest_production_stack.sh stop     # when done
```

Credentials: `workspace/auth.pentest.local` (generated on first start). ZAP target is usually `http://<lan-ip>/` (benserver example: `http://192.168.204.18/`). If LAN clients cannot reach :80, run `sudo ./scripts/open_lan_port.sh 80`.

See also Health checks (developer) for probe details and ZAP baseline expectations for accepted Medium/Low findings.

## 12.5 Docs preview

```
cd docs && bundle install && bundle exec jekyll serve
# http://127.0.0.1:4000/open-fdd/
```

# 13 Build Docker images

# 14 Build Docker images

## 14.1 Local build

```
./scripts/docker_build.sh
# Optional tar for air-gap:
./scripts/docker_build.sh --save
```

Images: `openfdd-bridge`, `openfdd-commission`, `openfdd-mcp-rag`.

## 14.2 Publish to GHCR (maintainers)

GitHub Actions workflow **Publish Docker addons** — manual dispatch; publishes **`:latest`** only and prunes GHCR (retired `openfdd-bacnet-poll` removed; keep 3 versions per image).

Maintainer checklist:

1. Green CI on `master`
2. Run **Publish Docker addons** workflow
3. Verify `ghcr.io/bbartling/openfdd-{bridge,commission,mcp-rag}:latest`
4. Edge hosts: `docker compose pull && docker compose up -d`

Details live in `.github/workflows/` and `scripts/docker_build.sh` — not duplicated here.

# 15 Developer Guide

# 16 Developer Guide

Clone, build, test, and contribute to Open-FDD.

| Topic | Page |
|---|---|
| Local setup | Local development |
| Docker builds | Build Docker images |
| Health checks | Health checks (developer) |

| Topic | Page |
|---|---|
| Tests & CI | Tests and CI |
| Security testing | Release & LAN scan cycle |
| PRs | Contributing |

AI-assisted contributors: see repository `AGENTS.md` for workspace layout, skill routing, and agent policy (one paragraph in public docs; details stay internal).

# 17 Tests and CI

# 18 Tests and CI

## 18.1 Local tests

```
./scripts/build_and_test.sh           # UI + workspace bridge pytest
python3 -m pytest open_fdd/tests -q   # PyPI package tests
cd workspace/dashboard && npm test    # dashboard unit tests (if configured)
```

## 18.2 CI (GitHub Actions)

| Workflow | Trigger | Checks |
|---|---|---|
| `ci.yml` | PR / push | pytest, bridge security audit, dashboard build, Jekyll docs |
| `docs-pages.yml` | push `master` | GitHub Pages site |
| `publish-open-fdd.yml` | tag `open-fdd-v*` | PyPI wheel |
| `publish-docker-addons.yml` | manual | GHCR images |

## 18.3 Docs build (same as CI)

```
cd docs && bundle exec jekyll build -d /tmp/open-fdd-site
```

Fix any template or link errors before opening a PR that touches `docs/`.

## 18.4 Security-related CI

`ci.yml` includes bridge security tests (`tests/workspace_bridge/test_security.py`). That complements — but does not replace — LAN ZAP + Nmap scans on each patch revision. See Security testing cycle.

# 19 Contributing

# 20 Contributing

## 20.1 Expectations

- Open an issue or discuss large changes before big refactors.
- Keep PRs focused; include test or docs updates when behavior changes.
- Do not commit secrets (`auth.env.local`, Ansible `secrets/*.env.local`, commissioning CSVs with real site data).
- Follow existing code style in `workspace/api/` and `workspace/dashboard/`.

## 20.2 Pull request checklist

- [ ] Tests pass locally (`./scripts/build_and_test.sh` or scoped pytest)
- [ ] Docs updated if user-facing behavior changed
- [ ] `docs/` Jekyll build succeeds
- [ ] No credentials in diff
- [ ] Auth/Caddy/header changes: re-run pentest verify + LAN ZAP and update ZAP baseline if expectations changed

## 20.3 Repository layout (short)

| Path | Contents |
|---|---|
| `workspace/api/` | Operator Bridge (FastAPI) |
| `workspace/dashboard/` | React SPA |
| `open_fdd/` | PyPI package (`arrow_runtime`, `playground`) |
| `docker/` | Compose files and image contexts |
| `infra/ansible/` | Edge deploy playbooks |
| `docs/` | This site (Jekyll + Just the Docs) |

## 20.4 Security issues

Do not file public issues for vulnerabilities. Contact the maintainer or use GitHub private security advisories.

## 20.5 Agent-assisted development

Cursor/Claude contributors: read `AGENTS.md` and `skills/` for automation boundaries. Public docs intentionally avoid agent playbook detail.

# 21 Health checks (developer)

# 22 Health checks (developer)

Extended validation for engineers with a **full git checkout** — CI, compose overlays, and pre-release gates.

IT operators on edge hosts should use Quick Start — health check and Deployment validation instead.

## 22.1 Clone and local stack

```
git clone https://github.com/bbartling/open-fdd.git && cd open-fdd
cp workspace/auth.env.example workspace/auth.env.local   # loopback dev only
./scripts/docker_build.sh                                # or: ./scripts/openfdd_stack.sh prod
./scripts/openfdd_stack.sh up
```

GHCR production pull (no local build):

```
OPENFDD_IMAGE_TAG=2026.06.07-edge ./scripts/openfdd_stack.sh prod
```

## 22.2 Stack health script

```
./scripts/stack_health_check.sh
OPENFDD_BASE_URL=http://192.168.204.18:8765 ./scripts/stack_health_check.sh
./scripts/stack_health_check.sh --require-ollama
```

Probes: containers up, `/health`, `/api/model/health`, sites, tree, SPARQL graph (with integrator auth when configured).

## 22.3 Compose overlays

| Overlay | Use |
|---|---|
| `docker/compose.dev.yml` | Local dev bind-mount |
| `docker/compose.bench.yml` | Host-network BACnet lab |
| `docker/compose.prod.yml` | GHCR images instead of local build |
| `docker/compose.edge.yml` | IT edge template (copy to `~/open-fdd/docker-compose.yml`) |

```
docker compose -f docker/compose.dev.yml -f docker/compose.bench.yml ps
docker compose -f docker/compose.dev.yml config --images
```

## 22.4 BACnet poll pipeline

```
tail -1 workspace/bacnet/polls/samples.csv
cat workspace/data/bacnet_ingest_state.json
docker compose logs --since 10m commission | grep -i poll
docker compose logs --since 10m bridge | grep -i ingest
```

## 22.5 pytest (bridge security / BACnet)

```
PYTHONPATH=workspace/api pytest tests/workspace_bridge/test_security.py -q
```

## 22.6 Pentest production stack (LAN ZAP)

```
./scripts/pentest_production_stack.sh start
./scripts/pentest_production_stack.sh verify
```

Then from a **LAN workstation**, run the packaged scan:

- Windows: `scripts/security/Run-OpenFddSecurityScan.ps1`
- macOS/Linux: `scripts/security/run_openfdd_security_scan.sh`

See scripts/security/README.md, Security — ZAP baseline, and the full security testing cycle. Host-side notes: `workspace/deploy/PENTEST.md`.

## 22.7 Post-deploy insurance (inventory hosts)

For automated checks against named inventory hosts, see `infra/ansible/scripts/post_deploy_check.sh` (maintainer tooling — not required for IT docker-pull sites).

# 23 Security testing cycle

# 24 Security testing cycle

How Open-FDD validates each **patch revision** before it ships to the live Acme HVAC bench — local automation, AI-assisted PR review, LAN passive scans, and edge post-deploy checks.

Related security docs: ZAP baseline, TLS and certificates, LAN hardening, Linux host hardening, Tenable remediation, Authenticated scanning (roadmap).

## 24.1 Who owns what

| Concern | Owner | Where |
|---|---|---|
| HTTP security headers (CSP, X-Frame, COOP, …) | **Bridge** middleware | `workspace/api/openfdd_bridge/security_headers.py` |
| TLS termination + HSTS | **Caddy** | |

| Concern | Owner | Where |
|---|---|---|
| | | `workspace/deploy/caddy/`, Ansible `Caddyfile.j2` |
| OT LAN TLS certificates | **Site integrator / maintainer** | `./scripts/setup_caddy_certs.sh` → `workspace/deploy/caddy/certs/`; Ansible `/etc/openfdd/caddy/` on edge |
| Pentest credentials | Generated per stack | `workspace/auth.pentest.local` (never commit) |
| LAN ZAP + Nmap scans | **Developer workstation** on test LAN | `scripts/security/` |

See TLS and certificate ownership for cert lifecycle.

# 24.2 Per-revision workflow

Typical cycle on a **control machine** (local dev) through a **lab edge host** (live bench):

```
flowchart LR
  A[Local dev + pytest] --> B[Pentest stack verify]
  B --> C[LAN ZAP + Nmap]
  C --> D[PR + GitHub CI]
  D --> E[CodeRabbit AI review]
  E --> F[Merge + tag open-fdd-vX.Y.Z]
  F --> G[GHCR + PyPI publish]
  G --> H[Acme upgrade + post_deploy_check --full]
  H --> I[Bump next patch locally]
```

## 24.2.1 1. Local functional tests

```
        ./scripts/build_and_test.sh
        PYTHONPATH=workspace/api pytest tests/workspace_bridge/
test_security.py -q
        infra/ansible/scripts/bench_5007_arrow_battery.sh   # when FDD/
Arrow paths change
```

### 24.2.2 2. Pentest production stack (on Open-FDD host)

```
./scripts/pentest_production_stack.sh start
./scripts/pentest_production_stack.sh verify
```

Confirms Caddy `:80` fronts the bridge, auth is enabled, BACnet writes disabled, and security headers are singular (no Caddy duplicates). Details: Health checks — pentest.

### 24.2.3 3. Host + exposure checks (on edge VM)

After IT Nessus scans or before Acme deploy sign-off:

```
./scripts/security/check_host_security.sh
./scripts/security/check_openfdd_exposure.sh --edge-ip <lan-ip>
```

Host remediation (dry-run first): `./scripts/security/remediate_ubuntu_host.sh`

### 24.2.4 4. LAN security scan (from laptop)

Use the packaged scripts — do not hand-roll one-off docker/nmap commands unless debugging:

| Platform | Command |
|---|---|
| Windows | `.\scripts\security\Run-OpenFddSecurityScan.ps1` |
| macOS / Linux | `./scripts/security/run_openfdd_security_scan.sh` |

Setup (Docker Desktop, Nmap): scripts/security/README.md.

Target URL is usually **`http://<lan-ip>/`** (Caddy), not `:8765`.

Review `openfdd-security-report/90-quick-findings-summary.txt` and compare against ZAP baseline expectations.

### 24.2.5 5. Pull request + automated review

- Open PR against `master`
- **GitHub Actions** (`ci.yml`): pytest, bridge security audit, dashboard build, Jekyll docs
- **CodeRabbit** (or similar AI PR review): style, security footguns, missing tests/docs

Do not merge with failing CI or unresolved duplicate-header regressions.

### 24.2.6 6. Release

```
git tag open-fdd-v3.0.4   # example
git push origin open-fdd-v3.0.4
```

Tag triggers GHCR `:latest` and PyPI publish workflows.

### 24.2.7 7. Lab edge validation

After pulling new images on a live bench VM:

```
OPENFDD_IMAGE_TAG=latest ./scripts/upgrade_edge_full.sh --limit
<inventory_host>
infra/ansible/scripts/post_deploy_check.sh --limit
<inventory_host> --full
```

Re-run LAN ZAP + Nmap against the edge LAN IP if Caddy or auth behavior changed. Site-specific example: GL36 lab note.

### 24.2.8 8. Next cycle

Bump patch version locally (`pyproject.toml`, `open_fdd/__init__.py`) on `master` for the next development round — do not tag until the next release is ready.

## 24.3 What each scan layer covers

| Layer | Tool | Depth |
|---|---|---|
| Unit / integration | pytest, `test_security.py` | Header middleware, auth gates |
| Stack smoke | `pentest_production_stack.sh verify` | Live headers, health, auth file |
| Passive web | ZAP baseline | Public routes, CSP, cookies, CORS |
| Port exposure | Nmap scoped | One host, known service ports |
| Authenticated API/dashboard | ZAP full + login | **Roadmap** — Authenticated scanning |

Baseline ZAP does **not** exercise integrator login or protected `/api/*` routes deeply.

## 24.4 Open-source contributors

The `scripts/security/` folder is part of the public repo so any fork can run the same LAN smoke against their bench. PRs that change Caddy, auth, or `security_headers.py` should update ZAP baseline and re-run the scan scripts.

# 25 Expression cookbook (Arrow-native)

# 26 Expression cookbook (Arrow-native)

Reference for **Open-FDD 3.x Rule Lab**: every rule is a Python module with **`apply_faults_arrow(table, cfg, context)`** using **`pyarrow.compute`** — **no pandas**, **no YAML expression files**, **no NumPy DataFrames on the IoT edge**.

This page **replaces the legacy pandas `type: expression` YAML cookbook**. Each legacy recipe maps to a **fixed fault code** and an Arrow pattern below.

| Topic | Page |
|---|---|
| Quick templates | Arrow recipes |
| Shared imports | Python recipes |
| Console window stats | Lookback window |
| Fault code catalog | Fault codes |
| Programmatic defaults | `open_fdd.arrow_runtime.sensor_catalog` |

---

## 26.1 Legacy → modern translation

| Legacy (pandas / YAML) | Arrow-native (3.x) |
|---|---|
| `type: expression` + `expression:` string | `apply_faults_arrow()` in `rules_py/*.py` |
| `RuleRunner.run(df, column_map=…)` | Rule Lab batch on **feather PyArrow table** |
| `np.maximum(a, b)` | `pc.max(a, b)` |
| `series.rolling(n).min()` | |

| Legacy (pandas / YAML) | Arrow-native (3.x) |
|---|---|
| | `arrow_rolling_min(vals, n)` from `open_fdd.arrow_runtime.windows` |
| `series.diff().abs()` | `arrow_abs_diff(vals, 1)` |
| `normalize_cmd(x)` | `pc.if_else(pc.greater(x, 1), pc.divide(x, 100), x)` |
| `params.max_temp` in YAML | Module constant or `cfg["bounds_high"]` |
| `flag: rule_a_flag` | Rule metadata **`fault_code`** → `AHU-A` … `VAV-C` |
| Numeric codes `VAV-03` | Letter codes **`VAV-C`** (see `LEGACY_CODE_MAP` in bridge) |

**Edge constraint:** Docker / BACnet poll path never imports pandas. Central portfolio Dash may use pandas for CSV analytics only.

---

# 26.2 How Arrow rules are structured

1. **Historian columns** — Feather column names from BACnet poll (`oa-t`, `sa-t`, `stat_zn-t`, …) or Brick labels via model export.
2. **Module constants** — Thresholds at top of file (bench style) or read from `cfg` for site tuning.
3. **`apply_faults_arrow`** — Returns a **boolean PyArrow array** (True = fault sample).
4. **`fault_code`** — Set in Rule Lab metadata; must exist in `GET /api/faults/catalog`.

```python
import pyarrow.compute as pc
from open_fdd.arrow_runtime.cookbook import sensor_bounds_mask

FAULT_CODE = "VAV-C"  # metadata in Rule Lab UI


def apply_faults_arrow(table, cfg, context=None):
    return sensor_bounds_mask(table, "zone_temp", cfg)
```

## 26.2.1 Command scaling (0–1 vs 0–100 %)

BACnet often exposes **0–100** for damper/VFD commands. Scale explicitly:

```python
def _norm_cmd(col):
    return pc.if_else(pc.greater(col, 1.0), pc.divide(col, 100.0), col)
```

### 26.2.2 Occupied hours (schedule gating)

Use `open_fdd.arrow_runtime.cookbook._unoccupied_mask` or compare local hour from `timestamp` column. Example: **fan on when unoccupied → BLD-C** / `schedule_compare`.

---

## 26.3 Sensor validation (bounds, flatline, rate of change)

Use **`open_fdd.arrow_runtime.sensor_catalog.SENSOR_PROFILES`** for defaults, or copy constants into `rules_py`. Tune per site in Rule Lab `cfg` or building-agent tuning brief.

### 26.3.1 Bounds (out of range) — `oob_rolling`

| Sensor kind | Min | Max | Flatline tol | Max Δ / hour | Max Δ / 15 min | Typical fault code |
|---|---|---|---|---|---|---|
| Zone temp (°F) | 55 | 90 | 0.10 | 4.0 | 2.0 | **VAV-C** |
| Supply air temp | 50 | 110 | 0.15 | 8.0 | 3.0 | **AHU-C**, **RTU-C** |
| Return air temp | 55 | 95 | 0.10 | 3.0 | 1.5 | **AHU-D** (also mixing) |
| Mixed air temp | 40 | 110 | 0.15 | 6.0 | 2.5 | **AHU-D** |
| Outdoor air temp | −40 | 130 | 0.10 | 12.0 | 6.0 | **BLD-B** |
| Duct static (in$H_2O$) | −0.5 | 3.0 | 0.02 | 0.5 | 0.25 | **AHU-A** |
| Relative humidity (%) | 0 | 100 | 1.0 | 15.0 | 8.0 | **DC-C** |
| Chilled water (°F) | 40 | 90 | 0.10 | 4.0 | 2.0 | **CH-D** |
| Hot water (°F) | 70 | 200 | 0.15 | 6.0 | 3.0 | **CH-D** |
| Condenser water (°F) | 50 | 110 | 0.15 | 5.0 | 2.5 | **CH-A** |
| $CO_2$ (ppm, occupied) | 400 | 1000 | 5.0 | 200 | 80 | **VAV-B** (ventilation) |

| Sensor kind | Min | Max | Flatline tol | Max Δ / hour | Max Δ / 15 min | Typical fault code |
|---|---|---|---|---|---|---|
| Discharge air temp | 45 | 120 | 0.15 | 10.0 | 4.0 | **VAV-E**, **HP-D** |

**Return air** uses a **narrow band** vs zone/OAT because RAT should track building return conditions, not outdoor extremes. When OAT/RAT/MAT are all present, prefer **mixing envelope** (below) over RAT bounds alone.

**$CO_2$ upper bound** is an occupied ventilation target; unoccupied rules may use 400–2000 ppm per policy.

### 26.3.2 Flatline (stuck sensor) — `flatline_1h`

Default **12 samples ≈ 1 hour** at 5-minute poll. True when rolling max − min ≤ tolerance.

```
        from open_fdd.arrow_runtime.cookbook import sensor_flatline_mask

        def apply_faults_arrow(table, cfg, context=None):
            return sensor_flatline_mask(table, "outdoor_air_temp", cfg)
# BLD-B
```

Bench references: `workspace/data/rules_py/bench_oa-t_flatline_1h.py`, `bench_stat_zn-t_flatline_1h.py`.

### 26.3.3 Rate of change (spike) — `rate_of_change`

Flags physically impossible step changes (comms glitch, substituted value).

```
        from open_fdd.arrow_runtime.cookbook import rate_of_change_mask
        from open_fdd.arrow_runtime.sensor_catalog import cfg_from_profile

        def apply_faults_arrow(table, cfg, context=None):
            merged = cfg_from_profile("outdoor_air_temp", cfg)
            merged["samples_per_hour"] = 12  # 5-min poll
            return rate_of_change_mask(table, merged, col="oa-t")
```

Legacy **weather_temp_spike** (`spike_limit: 16` °F per step) → **BLD-B** with `max_per_15min: 6` or stricter per-step limit.

### 26.3.4 Mixing envelope (MAT vs OAT/RAT) — `mixing_envelope`

Legacy GL36 **Rule B / C** and **AHU-D**. MAT should sit between OAT and RAT (± tolerance) while fan runs.

```python
from open_fdd.arrow_runtime.cookbook import mixing_envelope_mask

def apply_faults_arrow(table, cfg, context=None):
    return mixing_envelope_mask(
        table,
        {**cfg, "mixing_tol": 1.15},
        mat_col="ma-t",
        oat_col="oa-t",
        rat_col="ra-t",
        fan_col="supply-fan-speed-command",
    )
```

---

## 26.4 GL36-inspired AHU rules (legacy expression → fault code)

ASHRAE Guideline 36-style rules from the old YAML cookbook, translated to Arrow. Assign `fault_code` per row in Rule Lab.

| Legacy rule | Summary | Suggested code | Pattern |
|---|---|---|---|
| Rule A — duct static low @ full fan | SP below SP setpoint at high VFD | **AHU-A** | `custom_arrow` |
| Rule B — blend below band | MAT below OAT/RAT envelope | **AHU-D** | `mixing_envelope` |
| Rule C — blend above band | MAT above OAT/RAT envelope | **AHU-D** | `mixing_envelope` |
| Rule D — discharge cold when heating | SAT low vs MAT, heat valve open | **AHU-B** | `custom_arrow` |
| Rule E — SAT low, full heating | SAT below SP, valve > 90% | **AHU-C** / performance | `custom_arrow` |
| Rule F — SAT/ MAT | Econ mode, | **AHU-E** | `custom_arrow` |

| Legacy rule | Summary | Suggested code | Pattern |
|---|---|---|---|
| mismatch econ | SAT ≠ MAT | | |
| Rule G — ambient warm free cool | OAT > SAT SP, econ open, cool off | **AHU-E** | `custom_arrow` |
| Rule H — OAT/MAT mismatch econ+mech | Mech + econ, MAT ≠ OAT | **AHU-E** | `mixing_envelope` |
| Rule I — OAT/ MAT mismatch econ-only | Econ only, MAT ≠ OAT | **AHU-E** | `mixing_envelope` |
| Rule J — discharge above blend cooling | SAT > MAT in cooling | **AHU-B** | `custom_arrow` |
| Rule K — discharge above SP full cool | SAT > SP, full cooling | **AHU-C** | `custom_arrow` |
| Rule L — cooling coil ΔT when off | CHW drop when valves closed | **CH-C** | `custom_arrow` |
| Rule M — heating coil ΔT when off | HW rise when valves closed | **AHU-B** | `custom_arrow` |

### 26.4.1 Rule A — duct static low at full speed (Arrow sketch)

```
import pyarrow.compute as pc

SP = "duct-static-pressure"
SP_SP = "duct-static-pressure-sp"
FAN = "supply-fan-speed-command"
SP_MARGIN = 0.12
DRV_HI = 0.93


def apply_faults_arrow(table, cfg, context=None):
    sp = pc.cast(table[SP], "float64")
    sp_sp = pc.cast(table[SP_SP], "float64")
```

```
            fan = pc.if_else(pc.greater(table[FAN], 1),
pc.divide(table[FAN], 100), table[FAN])
            low_sp = pc.less(sp, pc.subtract(sp_sp, SP_MARGIN))
            fan_hi = pc.greater_equal(fan, DRV_HI - 0.06)
            return pc.and_(low_sp, fan_hi)
```

---

# 26.5 Starter pack (VAV / AHU baseline)

Maps to legacy YAML starter filenames; use as first deploy bundle.

| Legacy YAML starter | Arrow approach | Fau cod |
|---|---|---|
| `01_vav_zone_temp_bounds_occupied` | `sensor_bounds_mask(..., "zone_temp")` + occupied mask | **VAV C** |
| `02_vav_zone_temp_flatline_occupied` | `sensor_flatline_mask(..., "zone_temp")` + occupied | **VAV C** |
| `03_vav_damper_command_extreme_flatline` | `flatline_1h_mask` on damper cmd column | **VAV D** |
| `04_ahu_runtime_outside_schedule` | `after_hours_fan_satisfied_mask` / run-hours script | **BL C** |
| `05_ahu_duct_static_pressure_not_maintained` | Rule A sketch above | **AH A** |
| `06_ahu_internal_temp_sensor_bounds` | `sensor_bounds_mask` on SAT/MAT | **AH C** |
| `07_ahu_internal_temp_sensor_flatline` | `sensor_flatline_mask` on SAT | **AH C** |

## 26.5.1 Economizer starters

| Legacy name | Fault code | Key logic |
|---|---|---|
| `ahu_econ_100oa_temp_tracking_fault` | **AHU-E** | OA damper ≥ 95%, \|MAT−OAT\| and \|SAT−MAT\| > tol |
| `ahu_mech_cooling_when_free_cooling_available` | **AHU-E** | OAT ≤ SAT SP − margin, low OA damper, cooling valve on |

| Legacy name | Fault code | Key logic |
|---|---|---|
| `ahu_oa_damper_excess_open_extreme_ambient` | **AHU-E** | OA ≥ 50% when OAT > 70 °F or OAT < 40 °F |

## 26.6 VAV, central plant, heat pump

| Legacy recipe | Fault code | Notes |
|---|---|---|
| `zone_reheat_warm_ambient` | **VAV-A** | Reheat open when OAT > cutoff |
| `zone_damper_valve_full_open` | **VAV-D** | Damper > 97.5% for rolling window |
| `dp_below_sp_pump_max` | **CH-E** | Plant DP low at max pump speed |
| `flow_high_pump_max` | **CH-B** | High flow at max pump |
| `plant_supply_temp_deadband` | **CH-D** | CHW SP ± band while pump on |
| `chiller_excessive_runtime` | **CH-F** | Rolling sum of chiller on samples |
| `hp_discharge_cold_when_heating` | **HP-D** | SAT low, zone cold, fan on |

Rolling persistence in pandas `.rolling(n).min()` → `arrow_rolling_min` + `pc.and_` with current sample.

## 26.7 Opportunistic / ventilation

| Legacy recipe | Fault code |
|---|---|
| `econ_active_warm_ambient` | **AHU-E** |
| `mech_cool_when_econ_available` | **AHU-E** |
| `low_oa_fraction_estimated` | **VAV-B** / **BLD-A** |
| `preheat_excess_temp` | **AHU-B** |
| `weather_temp_spike` | **BLD-B** (`rate_of_change`) |
| `weather_gust_lt_wind` | **BLD-B** (custom compare) |

**OA fraction estimate** (no airflow meter):

```
# Guard: |RAT - OAT| > gap before dividing
oa_frac = (mat - rat) / (oat - rat) * 100
# Flag when oa_frac < oa_min_pct and fan on
```

Use `pc.and_` with gap check; skip divide where denominator near zero.

---

## 26.8 Data quality vs equipment faults

| Symptom | Pattern | Code | Do not confuse with |
|---|---|---|---|
| Flatline 1 h | `flatline_1h` | **VAV-C**, **AHU-C**, **BLD-B** | Equipment off / damper closed |
| OOB sample | `oob_rolling` | sensor-specific | True process excursion |
| Spike between polls | `rate_of_change` | **BLD-B** | Fast legitimate weather front |
| No new samples | `stale_points` | **BLD-D** | Equipment fault |
| MAT ∉ [OAT, RAT] | `mixing_envelope` | **AHU-D** | Stratification during startup |

Always gate sensor faults with **fan on**, **valve open**, or **occupied** where appropriate — see Sensor & data quality.

---

## 26.9 Metric (°C) equivalents

Set `cfg["temp_unit"] = "metric"` in Rule Lab or scale constants:

| Imperial | Metric (approx) |
|---|---|
| 55 °F | 12.8 °C |
| 90 °F | 32.2 °C |
| 50 °F | 10.0 °C |
| 110 °F | 43.3 °C |
| 2.0 °F tol | 1.1 °C |
| 0.15 in$H_2O$ | ~37 Pa |

`open_fdd.playground.temp_units` documents UI field keys for °F/°C toggle.

---

## 26.10 Binding rules to the fault catalog

1. Pick a **letter code** from Fault codes — never invent suffixes.
2. In Rule Lab, set **fault_code** on the rule row.
3. Batch FDD aggregates into `GET /api/faults/status` and portfolio rollup.
4. Legacy numeric codes (`AHU-03`) migrate via `LEGACY_CODE_MAP` in the bridge.

```
curl -s http://127.0.0.1:8765/api/faults/catalog | jq '.families[].codes[] | select(.code=="VAV-C")'
```

---

## 26.11 Pandas YAML → Arrow checklist (commissioning)

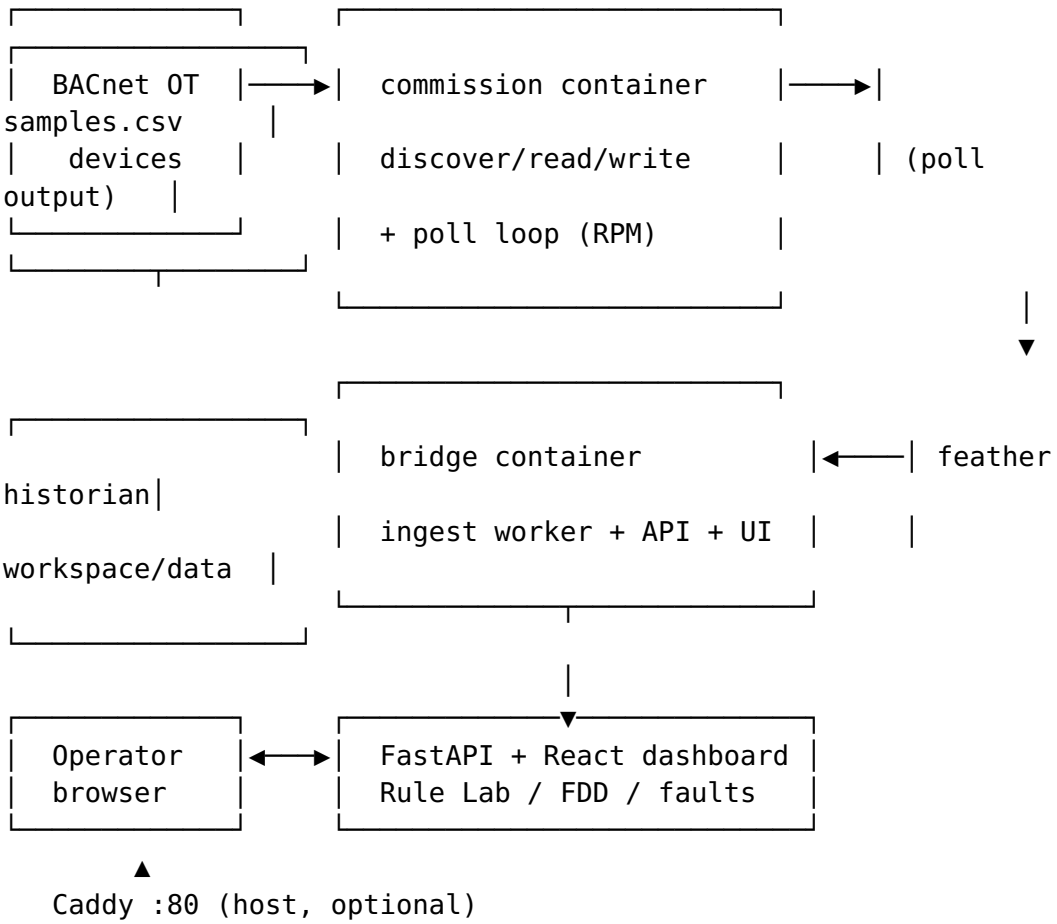
- [ ] Replace each `type: expression` YAML with a `rules_py` module
- [ ] Map `inputs` Brick labels → feather column names in model/poll CSV
- [ ] Set `fault_code` per rule (letter suffix)
- [ ] Run Rule Lab kit on 7-day feather window; confirm flag rate sane
- [ ] Apply sensor_catalog defaults; tune bounds for site climate

Enable building-agent check-in; verify faults in portfolio rollup

**Next:** Arrow recipes · Fault codes · Rule Lab

# 27 System overview

# 28 System overview




## 28.1 Docker edge stack (production)

Three GHCR images — see Containers:

| Container | BACnet OT | Operator LAN |
|---|---|---|
| **bridge** | No (HTTP only) | Yes, via Caddy `:80` or loopback `:8765` |
| **commission** | Yes (host network, `:47808`) | No (internal `:8767`) |

| Container | BACnet OT | Operator LAN |
|---|---|---|
| **mcp-rag** | No | No (internal `:8090`) |

BACnet **polling** is not a separate container — it runs inside **commission**. The bridge only **ingests** poll CSV into feather.

## 28.2 Components

| Layer | Responsibility |
|---|---|
| **Operator Bridge** | Auth, REST/WebSocket API, compiled dashboard, Rule Lab, historian ingest |
| **BACnet commission** | Who-Is, read/write (gated), **scheduled poll loop** → `samples.csv` |
| **Historian** | Feather files per point; local-first retention |
| **FDD runner** | `workspace/data/rules_py/*.py` on timer or `POST /api/rules/batch` |
| **Model** | Brick TTL + `model.json` bindings (`fdd_input`, equipment tree) |
| **MCP RAG** | Optional doc retrieval for agent tools |
| **Caddy** | Host reverse proxy `:80` → bridge (typical edge) |
| **Ollama** | Optional host LLM for building insight (not required for FDD) |

## 28.3 Deployment

IT operators: Quick Start — Docker. Modes and Caddy: Deployment modes.

## 28.4 Optional PyPI-only path

`pip install open-fdd` ships Arrow runtime + playground lint helpers **without** the Operator Bridge UI. Use for offline rule tests or CI. See Appendix — Python package.

# 29 Arrow recipes

# 30 Arrow recipes (default)

Open-FDD 3.0 Rule Lab rules use **`apply_faults_arrow(table, cfg, context)`**, **`pyarrow.compute`**, and **module constants** (no `config.json`).

For the full legacy pandas/YAML → Arrow translation, sensor bound tables, and GL36 fault-code mapping, see **Expression cookbook (Arrow-native)**.

## 30.1 Simple threshold

```
import pyarrow.compute as pc

VALUE_COLUMN = "zone_temp"
MAX_ZONE_TEMP = 78.0


def apply_faults_arrow(table, cfg, context=None):
    return pc.greater(table[VALUE_COLUMN], MAX_ZONE_TEMP)
```

## 30.2 Supply air temp high

```
import pyarrow.compute as pc

VALUE_COLUMN = "sa-t"
HIGH_F = 75.0


def apply_faults_arrow(table, cfg, context=None):
    return pc.greater(pc.cast(table[VALUE_COLUMN], "float64"),
HIGH_F)
```

## 30.3 Fan commanded but no airflow

```
import pyarrow.compute as pc

FAN_CMD = "fan-cmd"
AIRFLOW = "supply-cfm"
FAN_ON = 0.5
MIN_CFM = 500.0
```

```
        def apply_faults_arrow(table, cfg, context=None):
            fan_on = pc.greater(table[FAN_CMD], FAN_ON)
            low_airflow = pc.less(table[AIRFLOW], MIN_CFM)
            return pc.and_(fan_on, low_airflow)
```

## 30.4 Flatline (1 h rolling window)

```
        import pyarrow.compute as pc
        from open_fdd.arrow_runtime.windows import arrow_rolling_max,
arrow_rolling_min

        VALUE_COLUMN = "oa-t"
        WINDOW_SAMPLES = 12
        FLATLINE_TOLERANCE = 0.1


        def apply_faults_arrow(table, cfg, context=None):
            vals = pc.cast(table[VALUE_COLUMN], "float64")
            spread = pc.subtract(arrow_rolling_max(vals, WINDOW_SAMPLES),
arrow_rolling_min(vals, WINDOW_SAMPLES))
            return pc.less_equal(pc.abs(spread), FLATLINE_TOLERANCE)
```

## 30.5 Sensor out of range (catalog defaults)

```
        from open_fdd.arrow_runtime.cookbook import sensor_bounds_mask

        def apply_faults_arrow(table, cfg, context=None):
            return sensor_bounds_mask(table, "outdoor_air_temp", cfg)  #
fault_code: BLD-B
```

## 30.6 Sensor flatline + spike (catalog)

```
        from open_fdd.arrow_runtime.cookbook import sensor_flatline_mask,
rate_of_change_mask
        from open_fdd.arrow_runtime.sensor_catalog import cfg_from_profile

        def apply_faults_arrow(table, cfg, context=None):
            merged = cfg_from_profile("zone_temp", cfg)
            merged["samples_per_hour"] = 12
            flat = sensor_flatline_mask(table, "zone_temp", cfg)
            spike = rate_of_change_mask(table, merged, col="stat_zn-t")
            return pc.or_(flat, spike)  # import pyarrow.compute as pc
```

Bounds table: Expression cookbook — sensor validation.

More templates ship in `open_fdd.playground.arrow_templates` and via `GET /api/playground/arrow-templates`.

Bench examples: `workspace/data/rules_py/bench_*.py`.

Dev kit zip layout: Rule Lab — dev kit zip.

# 31 Deployment modes

# 32 Deployment modes

How Open-FDD is typically run: local development, lab LAN trials, and production edge on a trusted OT/IT network. Open-FDD is designed for **building LAN or OT edge** deployment — not direct exposure to the public internet.

## 32.1 Mode summary

| Mode | Stack | BACnet | Front door | Auth |
|---|---|---|---|---|
| **Local dev** | Compose on control machine | Optional bench simulator | `127.0.0.1:8765` | `OFDD_AUTH_DISABLED=1` optional on loopback |
| **Lab LAN** | Three GHCR containers | Real or simulator VLAN | Caddy `:80` or direct `:8765` | Required unless explicit insecure lab flags |
| **Edge production** | Three GHCR containers + host Caddy | OT NIC via **commission** | **Caddy** `:80` or `:443` → loopback bridge | **Required** — `workspace/auth.env.local` |

## 32.2 v3 container stack (all edge modes)

| Image | Role |
|---|---|
| `openfdd-bridge` | Operator Bridge API, dashboard, **feather historian ingest** |
| `openfdd-commission` | BACnet discover/read/write + **poll loop** |
| `openfdd-mcp-rag` | |

| Image | Role |
|---|---|
| | Doc-search sidecar for agent tools |

> **`openfdd-bacnet-poll` is retired.** BACnet polling runs inside **commission** only. Do not run the legacy poll container or `openfdd-bacnet-poll` systemd unit alongside Docker commission — that double-polls the OT network.

See Containers for ports, networks, and persistence.

## 32.3 Caddy HTTP (typical edge)

- Caddy listens on building LAN `:80`
- Reverse-proxies to bridge at `127.0.0.1:8765`
- Bridge stays loopback-only; operators bookmark `http://<edge-ip>/`

Bootstrap and compose: Quick Start — Docker.

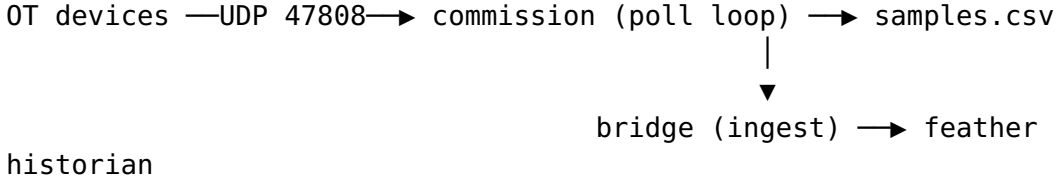
## 32.4 Caddy TLS (OT LAN)

For HTTPS on the edge LAN (self-signed or site CA):

1. Generate certs: `./scripts/setup_caddy_certs.sh` (control machine) or site PKI
2. Start with `OFDD_CADDY_MODE=tls`
3. Trust the CA on operator PCs

Details: TLS and certificates. Security headers are set by the **Operator Bridge**; Caddy adds **HSTS** only in TLS mode.

## 32.5 BACnet data path




The bridge thread named `bacnet_poll_worker` is **ingest only** — it does not bind BACnet.

## 32.6 What not to do on production edges

- Port-forward `:8765` to the public internet without TLS and strong auth
- Run `docker compose down -v` or delete `workspace/` on a live site
- Deploy the retired `openfdd-bacnet-poll` image alongside commission

## 32.7 Related

| Topic | Page |
|---|---|
| First deploy | Quick Start — Docker |
| Upgrade | Updating the stack |
| LAN security | LAN hardening |
| Lab example site | Examples — GL36 lab note |

# 33 Python recipes (Arrow)

# 34 Python recipes (Arrow)

All Rule Lab Python rules use **`apply_faults_arrow(table, cfg, context)`** on PyArrow tables with **`pyarrow.compute`** and **module constants**.

See **Arrow recipes** for the canonical cookbook (flatline, spread, OOB, schedule faults).

Shared helpers live in `open_fdd.arrow_runtime.cookbook` and `open_fdd.arrow_runtime.windows`. For console validation, copy **`_kit_lookback_stats`** from Lookback window helper.

```
        from open_fdd.arrow_runtime.windows import arrow_rolling_min,
arrow_rolling_max
        import pyarrow.compute as pc

        VALUE_COLUMN = "oa-t"
        WINDOW_SAMPLES = 12
        FLATLINE_TOLERANCE = 0.1

        def apply_faults_arrow(table, cfg, context=None):
            vals = pc.cast(table[VALUE_COLUMN], "float64")
            spread = pc.subtract(arrow_rolling_max(vals, WINDOW_SAMPLES),
```

```
arrow_rolling_min(vals, WINDOW_SAMPLES))
            return pc.less_equal(pc.abs(spread), FLATLINE_TOLERANCE)
```

# 35 Containers

# 36 Containers

Open-FDD v3 edge sites run **three** GHCR containers plus optional **host** services (Caddy, Ollama). There is no separate BACnet poll container — **field BACnet polling lives in commission**.

**Quick answer:** Who binds BACnet UDP 47808? **`openfdd-commission`**. Bridge only ingests `samples.csv` into the feather historian.

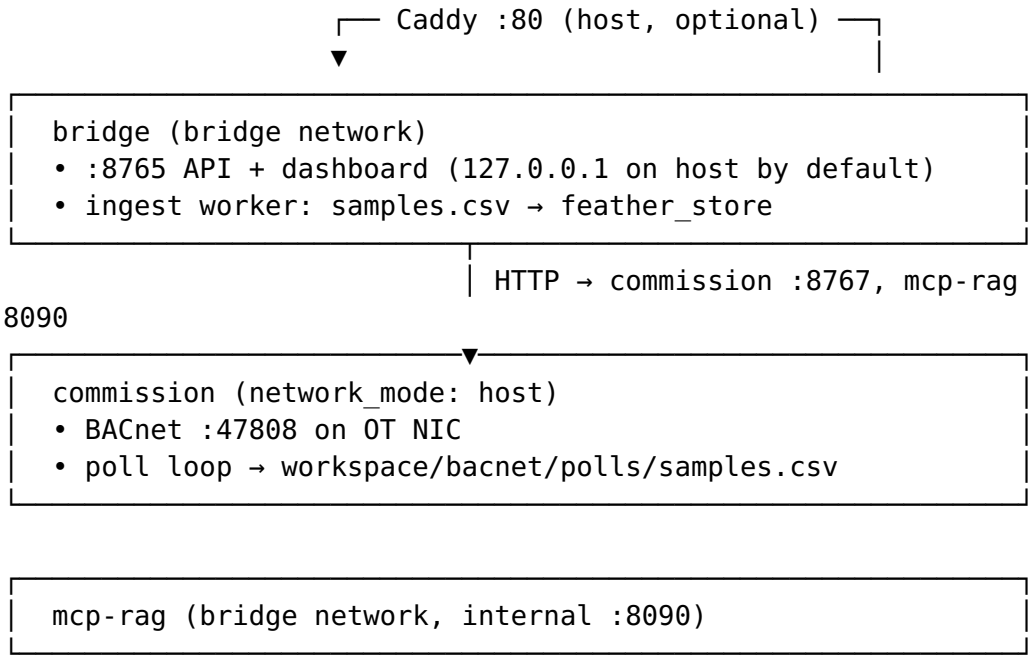
## 36.1 GHCR images

| Image | Compose service | Role |
|---|---|---|
| `ghcr.io/bbartling/openfdd-bridge` | `bridge` | Operator API, React dashboard, feather **ingest** |
| `ghcr.io/bbartling/openfdd-commission` | `commission` | BACnet discover / read / write + **poll loop** |
| `ghcr.io/bbartling/openfdd-mcp-rag` | `mcp-rag` | Doc-search sidecar for agent tools |

Tag: **`latest`** on GitHub Packages. Retired: `openfdd-bacnet-poll` (poll runs inside **commission**). GHCR retains only the **three newest** versions per image.

Template: `docker/compose.edge.yml` → copy to `~/open-fdd/docker-compose.yml`.

## 36.2 Stack diagram




## 36.3 Responsibilities

| Concern | Container |
|---|---|
| BACnet Who-Is / read / write | **commission** |
| BACnet scheduled RPM reads | **commission** (`bacnet_poll_loop`) |
| CSV → feather historian | **bridge** (feather ingest worker — `bacnet_poll_worker` module name is historical) |
| Operator login, Rule Lab, faults UI | **bridge** |
| Agent doc search | **mcp-rag** |

The bridge thread named `bacnet_poll_worker` is **ingest only** — it does not bind BACnet.

## 36.4 Network and ports (typical edge)

| Endpoint | Reachable from | Notes |
|---|---|---|
| Caddy `:80` | Building LAN | Reverse proxy to bridge |
| Bridge `:8765` | Loopback (default) | |

| Endpoint | Reachable from | Notes |
|---|---|---|
| | | Bind `127.0.0.1` in compose |
| Commission : `8767` | Bridge / loopback | HTTP API for BACnet ops |
| Commission : `47808` | OT BACnet LAN | UDP; requires `network_mode: host` |
| MCP `:8090` | Bridge container network | Not exposed to LAN by default |

## 36.5 Long-lived deployment

All compose services set `restart: unless-stopped`. After host reboot:

```
sudo systemctl enable docker
cd ~/open-fdd && docker compose ps
```

If containers are down: `docker compose up -d` (idempotent).

See Run with Docker images.

## 36.6 Persistent data

State lives on the **host** under `workspace/` (bind mount). Containers are replaceable; **backup `workspace/`** before image upgrades.

| Path | Contents |
|---|---|
| `workspace/data/feather_store/` | Historian |
| `workspace/data/*.json` | Model, rules, FDD results |
| `workspace/bacnet/commissioning/` | `commission.env`, `points.csv` |
| `workspace/bacnet/polls/samples.csv` | Poll output |
| `workspace/auth.env.local` | Login secrets |
| `workspace/api/static/app/` | Dashboard bundle (if updated separately) |

## 36.7 Optional host services

| Service | Role |
|---|---|
| **Caddy** | TLS or plain HTTP `:80` → `127.0.0.1:8765` |
| **Ollama** | Optional LLM on `:11434` (not in the three-image stack) |
| **FDD loop timer** | Optional `openfdd-fdd-loop.timer` on host (`docker exec` batch) |

# 37 Lookback window helper

# 38 Lookback window helper

Rules that depend on **time history** (flatline, spread, streaks, trim/respond validation) should print a clear window summary in the Rule Lab console or local `run_test.py` output.

## 38.1 Recommended pattern (copy into `rule.py`)

Use module constants for the lookback you intend; the bridge passes the full feather window into `apply_faults_arrow` based on batch `lookback_hours` (default **1 h** on the scheduled loop; **24 h** when using **Update all records** with chunks).

```
        import pyarrow.compute as pc

        LOOKBACK_HOURS = 3  # tune: 1, 3, 6, 12, 24


        def _kit_lookback_stats(table, *, hours=None):
            """Dev helper — prints row count, timestamp span, and
configured lookback."""
            h = hours if hours is not None else LOOKBACK_HOURS
            if "timestamp" not in table.column_names:
                print(f"lookback={h}h rows={table.num_rows} (no timestamp
column)")
                return
            ts = pc.cast(table["timestamp"], "timestamp[us, UTC]")
            tmin = pc.min(ts).as_py()
```

```
            tmax = pc.max(ts).as_py()
            if tmin is None or tmax is None:
                print(f"lookback={h}h rows={table.num_rows} start=None
stop=None span=0.00h")
                return
            span_h = (tmax - tmin).total_seconds() / 3600.0
            print(
                f"lookback={h}h rows={table.num_rows} "
                f"start={tmin.isoformat()} stop={tmax.isoformat()}
span={span_h:.2f}h"
            )


        def _kit_value_stats(table, column):
            vals = pc.cast(table[column], "float64")
            print(
                f"column={column} min={pc.min(vals).as_py():.2f} "
                f"max={pc.max(vals).as_py():.2f}
mean={pc.mean(vals).as_py():.2f}"
            )


        def apply_faults_arrow(table, cfg, context=None):
            _kit_lookback_stats(table)
            _kit_value_stats(table, "oa-t")
            # ... fault logic ...
```

# 38.2 Why not hide this behind a library call yet?

A future `open_fdd.arrow_runtime.kit.lookback_stats(table, hours=3)` helper is reasonable once the Algorithms tab shares the same kit export path. For now, keeping the helper **inline in `rule.py`** matches the constants-first Rule Lab style and prints the same fields in:

- Local `run_test.py`
- Rule Lab **Quick test** console
- Batch run logs

# 38.3 Lookback vs rolling samples

| Concept | Controlled by | Typical use |
|---|---|---|
| **Historian lookback** | Batch `lookback_hours`, FDD loop env | How many rows are in `table` |

| Concept | Controlled by | Typical use |
|---|---|---|
| **Rolling window** | `WINDOW_SAMPLES` constant (~12 ≈ 1 h @ 5 min poll) | Flatline, spread, consecutive-true |

Always call `_kit_lookback_stats(table)` first so operators can confirm the table span matches expectations before interpreting rolling-window faults.

## 38.4 Batch intervals

| Trigger | Default lookback |
|---|---|
| `openfdd-fdd-loop` / `OFDD_FDD_INTERVAL_MINUTES` | 1 h |
| Rule Lab **Update all records** | 24 h (`chunk_hours: 6`, `use_chunks: true`) |
| Rule Lab Quick test | 3 h (UI default) |

See Windowing & debugging for `arrow_rolling_min` / `arrow_consecutive_true`.

# 39 Data flow

# 40 Data flow

## 40.1 Telemetry ingest

Open-FDD supports three commissioning drivers. Each appends long-format samples to workspace CSVs and writes **feather shards** tagged by `source` (`bacnet`, `modbus`, `json_api`).

### 40.1.1 BACnet

1. **Commission poll loop** (inside the `commission` container) reads enabled BACnet points from `points.csv` on a schedule (host network, BACnet `:47808`).
2. RPM results append to `workspace/bacnet/polls/samples.csv`.
3. **Bridge ingest worker** loads new rows into `feather_store/bacnet/`.

```
BACnet devices → commission (poll loop) → samples.csv → bridge (ingest) → feather_store/bacnet
```

### 40.1.2 Modbus TCP

1. Operator configures registers on the **Modbus** tab (or `read_and_store` API).
2. Poll worker reads holding/input registers via Modbus TCP.
3. Samples append to `workspace/modbus/polls/samples.csv` → `feather_store/modbus/`.

### 40.1.3 JSON API (HTTP/HTTPS)

1. Operator configures REST endpoints on the **JSON API** tab — GET/POST, Bearer or Basic auth, `${ENV:VAR}` placeholders from `json_api.env.local`, optional TLS verify off for self-signed OT gateways.
2. Poll worker issues HTTP requests (one GET can fan out to multiple JSON paths, e.g. OpenWeather `web-oat-t` / `web-rh` / `web-weather-desc`).
3. Samples append to `workspace/json_api/polls/samples.csv` → `feather_store/json_api/`.
4. FDD batch **merges** `json_api` columns with BACnet/Modbus on the same site (nearest timestamp, 30 min tolerance) for cross-source rules.

```
OT REST / weather API → bridge JSON API worker → samples.csv →
feather_store/json_api
                                                        ↘ merged
into FDD frame with bacnet/modbus
```

Showcase: OpenWeatherMap bundle. Details: Driver framework, BACnet polling, JSON API driver, Containers.

## 40.2 Rule evaluation

1. Rules are Python files in `workspace/data/rules_py/` with metadata in `rules_store.json`.
2. **Bindings** map rules to points via the Brick model (`fdd_input` tags).
3. Batch runner (`POST /api/rules/batch` or host timer) produces `fdd_results.json`.
4. Dashboard **faults** view and check-engine status read aggregated results.

## 40.3 Model sync

- Commissioning imports update `model.json` / TTL graph.
- `POST /api/model/sync-ttl` refreshes SPARQL-backed tree used by the UI.

## 40.4 Retention

Feather files grow on disk; retention is configurable (`workspace/data.env.local`). Image upgrades must **preserve** `workspace/data/` — backup before `docker compose up -d --force-recreate`. See Updating the stack.

# 41 Architecture

# 42 Architecture

Open-FDD v3 edge: **three Docker containers** (`bridge`, `commission`, `mcp-rag`), BACnet polling in **commission**, feather ingest in **bridge**, optional host Caddy and Ollama.

| Page | Topic |
|---|---|
| System overview | Components and diagram |
| Deployment modes | Local dev, lab LAN, edge production, Caddy HTTP/TLS |
| Data flow | BACnet / Modbus / JSON API → feather → FDD → dashboard |
| Containers | GHCR images, ports, persistence, retired poll image |
| Driver framework | Shared commissioning pattern for OT sources |

**Operators:** Quick Start — Docker — no git clone on the edge host.

**BACnet polling:** Polling — commission container only.

# 43 Rule Cookbook

# 44 Rule Cookbook

Practical patterns for **Arrow-native Rule Lab** (`apply_faults_arrow`) on feather historian PyArrow tables — **no pandas on the IoT edge**.

| Page | Use when |
|---|---|
| **Expression cookbook (Arrow-native)** | **Full reference** — legacy YAML/pandas translation, sensor bounds, GL36, fault codes |
| Arrow recipes | Short templates — threshold, flatline, OOB, fan/schedule |
| Python recipes | Shared `open_fdd.arrow_runtime.cookbook` imports |
| Lookback window helper | `_kit_lookback_stats` — print start/stop timestamps and span |
| Windowing & debugging | Rolling windows, batch runtime, Rule Lab console |
| GL36 algorithm stubs | Doc-only supervisory patterns (trim & respond, plant resets) |

Programmatic sensor defaults: `open_fdd.arrow_runtime.sensor_catalog.SENSOR_PROFILES`.

```
        import pyarrow.compute as pc
        from open_fdd.arrow_runtime.cookbook import sensor_bounds_mask

        def apply_faults_arrow(table, cfg, context=None):
            return sensor_bounds_mask(table, "zone_temp", cfg)  # fault_code: VAV-C
```

Future graph ML (sklearn / PyG) runs in a separate training service — see GitHub issue #211.

# 45 Windowing & debugging

# 46 Windowing & debugging

## 46.1 Lookback vs rolling window

| | Historian lookback | Rolling window |
|---|---|---|
| Set by | Batch `lookback_hours`, FDD loop, Quick test UI | `WINDOW_SAMPLES` constant in `rule.py` |
| Typical | 1 h scheduled; 24 h Update all | 12 samples ≈ 1 h @ 5 min poll |
| Validate with | `_kit_lookback_stats(table)` — see Lookback window helper | Rule logic + console `flagged` count |

## 46.2 Rolling windows (Arrow)

Use `open_fdd.arrow_runtime.windows` for sample-based rolling min/max and consecutive-true streaks:

```
from open_fdd.arrow_runtime.windows import arrow_rolling_min, arrow_rolling_max, arrow_consecutive_true
```

Cookbook masks (`flatline_1h_mask`, `spread_1h_mask`, `oob_mask`) default to ~12 samples (~1 h at 5 min poll).

Bench rules use constants: `WINDOW_SAMPLES`, `FLATLINE_TOLERANCE`, `OAT_LOW` / `OAT_HIGH`, etc. Legacy `cfg` keys still work for uploaded rules that read `cfg.get(...)`.

## 46.3 Rule Lab console

Quick-test and batch responses include `backend: arrow`, `ms`, and flagged row counts. Arrow summary events appear in the event console on fault hits.

# 47 GL36 algorithm stubs

# 48 GL36 algorithm stubs (doc-only)

Draft **supervisory** PyArrow patterns for the future Algorithms tab. These mirror Niagara GL36 blocks documented in:

**README_TRIM_RESPOND.md**

FDD rules return **fault masks**. Algorithms return **setpoints or request integers** — the stubs below use `apply_algorithm_arrow` as the planned entrypoint. For early testing in Rule Lab, you can adapt the logic into advisory FDD rules (boolean “sequence unhealthy” flags).

## 48.1 Shared lookback helper

```
import pyarrow.compute as pc

LOOKBACK_HOURS = 6
```

```
        def _kit_lookback_stats(table, *, hours=None):
            h = hours if hours is not None else LOOKBACK_HOURS
            ts = pc.cast(table["timestamp"], "timestamp[us, UTC]")
            tmin = pc.min(ts).as_py()
            tmax = pc.max(ts).as_py()
            span_h = (tmax - tmin).total_seconds() / 3600.0 if tmin and
tmax else 0.0
            print(f"lookback={h}h rows={table.num_rows} start={tmin}
stop={tmax} span={span_h:.2f}h")
```

# 48.2 VAV zone cooling requests (0–3)

Simplified from GL36 VAV box request generator — zone temp vs cooling setpoint bands.

```
        """GL36 VAV cooling requests — doc stub."""

        import pyarrow.compute as pc

        ZONE_TEMP = "zn-t"
        ZONE_SP = "zn-t-sp"
        ZONE_DEMAND = "zn-cool-loop"
        HIGH_DIFF_F = 5.0
        MED_DIFF_F = 3.0
        LOOP_ON = 95.0
        LOOP_OFF = 85.0


        def apply_algorithm_arrow(table, cfg, context=None):
            _kit_lookback_stats(table)
            tz = pc.cast(table[ZONE_TEMP], "float64")
            sp = pc.cast(table[ZONE_SP], "float64")
            diff = pc.subtract(tz, sp)
            demand = pc.cast(table[ZONE_DEMAND], "float64")
            # Last row request level for kit demo (full port needs timer
state)
            last_diff = diff[-1].as_py() if table.num_rows else 0.0
            last_demand = demand[-1].as_py() if table.num_rows else 0.0
            if last_diff >= HIGH_DIFF_F:
                req = 3
            elif last_diff >= MED_DIFF_F:
                req = 2
            elif last_demand > LOOP_ON:
                req = 1
            else:
                req = 0
            print(f"cooling_requests={req} dT={last_diff:.1f}F
loop={last_demand:.0f}%")
            return {"cooling_requests": req}
```

# 48.3 AHU duct static trim & respond (advisory fault)

Detect **duct static stuck high** while VAV pressure requests are low — trim & respond may not be trimming.

```
        """AHU duct static trim advisory — doc stub."""

        import pyarrow.compute as pc

        DUCT_SP = "duct-static"
        DUCT_SP_MAX = 1.50
        DUCT_SP_TRIM_TARGET = 0.80
        VAV_PRESS_REQ_SUM = "vav-press-req-sum"
        LOOKBACK_HOURS = 12


        def apply_faults_arrow(table, cfg, context=None):
            _kit_lookback_stats(table, hours=LOOKBACK_HOURS)
            sp = pc.cast(table[DUCT_SP], "float64")
            reqs = pc.cast(table[VAV_PRESS_REQ_SUM], "float64")
            stuck_high = pc.and_(pc.greater(sp, DUCT_SP_TRIM_TARGET),
pc.less(reqs, 1.0))
            return pc.and_(stuck_high, pc.greater(sp, DUCT_SP_MAX * 0.9))
```

# 48.4 Chiller plant enable (valve request counting)

From GL36 §5.18.15.2 — count AHUs with CHW valve above threshold.

```
        """Chiller plant enable advisory — doc stub."""

        import pyarrow.compute as pc

        AHU_VALVE_COLS = ["ahu1-clg-vlv", "ahu2-clg-vlv", "ahu3-clg-vlv"]
        REQ_THRESH = 95.0
        DIS_THRESH = 10.0
        MIN_AHUS_REQ = 2


        def apply_algorithm_arrow(table, cfg, context=None):
            _kit_lookback_stats(table)
            requesting = 0
            for col in AHU_VALVE_COLS:
                if col not in table.column_names:
                    continue
                v = pc.cast(table[col], "float64")[-1].as_py()
                if v >= REQ_THRESH:
```

```
                requesting += 1
            enable = requesting >= MIN_AHUS_REQ
            print(f"chiller_enable={enable} requesting_ahus={requesting}/
{MIN_AHUS_REQ}")
            return {"chiller_enable": enable, "requesting_ahus":
requesting}
```

# 48.5 Hot water plant HWST trim & respond (advisory)

Flags **HWST trimmed too low** while heating requests remain high.

```
        """HWST trim advisory — doc stub."""

        import pyarrow.compute as pc

        HWST = "hw-supply-t"
        HWST_SP = "hw-supply-t-sp"
        HEAT_REQ_SUM = "hw-reset-req-sum"
        LOW_HWST_F = 120.0
        LOOKBACK_HOURS = 6


        def apply_faults_arrow(table, cfg, context=None):
            _kit_lookback_stats(table, hours=LOOKBACK_HOURS)
            temp = pc.cast(table[HWST], "float64")
            reqs = pc.cast(table[HEAT_REQ_SUM], "float64")
            return pc.and_(pc.less(temp, LOW_HWST_F), pc.greater(reqs,
2.0))
```

# 48.6 Chilled water plant (DP + CHWST reset advisory)

Flags **CHWST at design cold** with low cooling requests — possible stuck 100% plant loop.

```
        """CHW plant reset advisory — doc stub."""

        import pyarrow.compute as pc

        CHWST = "chw-supply-t"
        CHWST_DESIGN_COLD_F = 44.0
        COOL_REQ_SUM = "chw-reset-req-sum"
        LOOKBACK_HOURS = 6


        def apply_faults_arrow(table, cfg, context=None):
```

```
            _kit_lookback_stats(table, hours=LOOKBACK_HOURS)
            temp = pc.cast(table[CHWST], "float64")
            reqs = pc.cast(table[COOL_REQ_SUM], "float64")
            return pc.and_(pc.less(temp, CHWST_DESIGN_COLD_F + 1.0),
pc.less(reqs, 1.0))
```

## 48.7 Testing later

1. Bind historian columns in **Model & assignments** (`fdd_input` / brick types).
2. Copy a stub into Rule Lab as an FDD advisory rule, or wait for the **Algorithms** tab kit export.
3. Run `python run_test.py` from a downloaded kit and confirm `_kit_lookback_stats` prints expected `start` / `stop` / `span`.

Reference implementation (Niagara Java): README_TRIM_RESPOND.md.

# 49 Central plants (CHW / CTW / BLR)

# 50 Central plant Arrow recipes

Chiller plant, condenser water, cooling tower, boiler, and hot-water distribution faults use the same Arrow-native primitives as air-side rules.

## 50.1 Starter fault codes

| Code | Primitive | Recipe |
|---|---|---|
| `CHW-DT-001` | `low_delta_t_mask` | Low chilled-water ΔT syndrome |
| `CHW-DP-001` | `reset_stuck_mask` | CHW differential pressure reset stuck high |

## 50.2 Required point roles (CHW)

- `chw_supply_temp`, `chw_return_temp` — ΔT rules
- `chw_dp`, `chw_dp_setpoint` — reset stuck rules
- `pump_command`, `pump_feedback` — command/feedback mismatch

## 50.3 Synthetic test pattern

```python
import pyarrow as pa
from open_fdd.arrow_runtime.primitives import low_delta_t_mask

table = pa.table({
    "chw_supply_temp": [44.0] * 10,
    "chw_return_temp": [44.5] * 10,
})
mask = low_delta_t_mask(
    table,
    {"min_delta_t": 4.0},
    supply_col="chw_supply_temp",
    return_col="chw_return_temp",
)
assert mask.to_pylist()[-1] is True
```

See Fault codes — chiller plant (expanded in 3.0.19).

# 51 Dashboard overview

# 52 Dashboard overview

## 52.1 Primary areas

| Area | Purpose |
|---|---|
| **Home / Building insight** | Check-engine status (green/yellow/red), active fault summary |
| **Trends** | Point history from feather store |
| **Faults** | Rule hits grouped by equipment family |
| **Rule Lab** | Edit, lint, and test Python FDD rules (PyArrow kit zip) |
| **Algorithms** | Supervisory GL36 trim & respond (coming soon) |

| Area | Purpose |
|---|---|
| **Model & assignments** | Equipment tree, FDD rule pins, commissioning JSON |
| **BACnet** | Discovery, reads, commission workflows |
| **Settings / health** | Stack status, auth mode |

## 52.2 Live updates

Dashboard tiles use REST polling and `WS /ws/dashboard` for selected live views (ticket auth via `POST /api/auth/ws-ticket`).

## 52.3 Roles

Read-only operators see trends and faults; integrators access Rule Lab and model import. See Auth and roles.

# 53 Auth and roles

# 54 Auth and roles

## 54.1 Production (LAN / edge)

- Set `OFDD_AUTH_SECRET` and role passwords in `workspace/auth.env.local`.
- Bridge on non-loopback addresses **requires** auth unless explicit insecure dev flags are set (lab only).

| Role | Typical access |
|---|---|
| `viewer` | Read trends, faults, model tree |
| `operator` | Run batch FDD, acknowledge views |
| `integrator` | Rule Lab, bindings, model import |
| `commission` | BACnet writes (plus write guard) |
| `agent` | Agent tools / app-edit gates |
| `admin` | Full configuration |

Login: `POST /api/auth/login` → Bearer JWT on API calls.

## 54.2 Localhost dev

`OFDD_BRIDGE_HOST=127.0.0.1` with `OFDD_AUTH_DISABLED=1` is acceptable on a single machine. **Never** use auth-disabled mode on a building LAN.

Full hardening: Security.

# 55 Rule Lab

# 56 Rule Lab

Rule Lab authors **Arrow-native** Python FDD rules against the feather historian. Bench rules use **module constants** at the top of `rule.py` — no browser config panel, no `config.json` in the dev kit zip.

```
        """Bench OA-T out of bounds (Arrow)."""

        import pyarrow.compute as pc

        VALUE_COLUMN = "oa-t"
        OAT_LOW = 68.0
        OAT_HIGH = 88.0
        LOOKBACK_HOURS = 1


        def _kit_lookback_stats(table, *, hours=None):
            h = hours if hours is not None else LOOKBACK_HOURS
            ts = pc.cast(table["timestamp"], "timestamp[us, UTC]")
            tmin = pc.min(ts).as_py()
            tmax = pc.max(ts).as_py()
            span_h = (tmax - tmin).total_seconds() / 3600.0 if tmin and
tmax else 0.0
            print(f"lookback={h}h rows={table.num_rows} start={tmin}
stop={tmax} span={span_h:.2f}h")


        def _kit_value_stats(table):
            vals = pc.cast(table[VALUE_COLUMN], "float64")
            print(
                f"column={VALUE_COLUMN} min={pc.min(vals).as_py():.2f} "
                f"max={pc.max(vals).as_py():.2f}
mean={pc.mean(vals).as_py():.2f}"
            )
```

```
        def apply_faults_arrow(table, cfg, context=None):
            _kit_lookback_stats(table)
            _kit_value_stats(table)
            vals = pc.cast(table[VALUE_COLUMN], "float64")
            return pc.or_(pc.less(vals, OAT_LOW), pc.greater(vals,
OAT_HIGH))
```

# 56.1 Workflow

1. **Download kit** — PyArrow zip (see below)
2. **Edit constants** locally (`VALUE_COLUMN`, limits, `LOOKBACK_HOURS`, `WINDOW_SAMPLES`)
3. **Run** `pip install -r requirements.txt` then `python run_test.py`
4. **Upload** `rule.py` on Rule Lab (integrator)

Upload validation (Phase B): AST parse, forbidden imports, `apply_faults_arrow(table, cfg, context=None)` signature.

The bridge runs rules on **PyArrow tables** via `open_fdd.arrow_runtime`, and persists `.py` sources under `workspace/data/rules_py/`.

- **Quick test** — `POST /api/playground/test-rule` (default 3 h lookback)
- **Batch** — `POST /api/rules/batch` / `openfdd-fdd-loop` timer (**1 h** default lookback)
- **Update all records** — 24 h lookback, 6 h chunks (`use_chunks: true`)
- **Templates** — `GET /api/playground/arrow-templates`

Pin points to rules via **Model & assignments** (`/model`) commissioning JSON or BACnet tree right-click. Export shows Rule Lab **names** on each point (`fdd_rules_linked`); import uses rule **ids** (`fdd_rule_ids`).

Bench seed: `python3 scripts/setup_bench_afdd.py` — imports `bench_import_model.json` and bench cookbook rules with matching id/name pairs.

# 56.2 Dev kit zip (download)

**Download kit** on Rule Lab exports `openfdd-rule-kit-<rule_id>.zip` with:

| File | Purpose |
|---|---|
| `rule.py` | Arrow rule + **constants** at top; optional `_kit_lookback_stats` / `_kit_value_stats` helpers |

| File | Purpose |
|---|---|
| `data.py` | `SITE_ID`, `LOOKBACK_HOURS`, `load_table()` reading `sample.feather` |
| `sample.feather` | Full historian window for the kit lookback (all rows, not a CSV preview) |
| `run_test.py` | Runs `rule.apply_faults_arrow(table, {}, context=...)` and prints `flagged=N` |
| `requirements.txt` | `open-fdd>=3.0.1` and `pyarrow` |
| `README.md` | Install and run instructions |
| `column_map.json` | BRICK logical keys → feather column names |
| `kit_meta.json` | Export metadata (site, columns, row count, lookback hours) |

**Not included:** `config.json` (retired — tune thresholds as Python constants).

Local test:

```
unzip openfdd-rule-kit-*.zip
cd openfdd-rule-kit-*
pip install -r requirements.txt
python run_test.py
```

Expected console output:

```
site=demo lookback_h=3 rows=…
lookback=3h rows=… start=… stop=… span=2.98h
column=oa-t min=… max=… mean=…
flagged=…
```

# 56.3 Lookback windows

Rules that need multi-hour history should call `_kit_lookback_stats(table)` (see Lookback window helper). Pass `hours=1`, `3`, `6`, `12`, or `24` to validate timestamp span in the console.

Rolling-window rules (flatline, spread) use `WINDOW_SAMPLES` (~12 samples ≈ 1 h at 5 min poll) — see Windowing & debugging.

## 56.4 Algorithms (supervisory — coming soon)

GL36 trim & respond and plant reset sequences will live on the **Algorithms** tab with the same zip kit shape but `apply_algorithm_arrow` outputs. Reference: README_TRIM_RESPOND.

## 56.5 Related

- Model workflow
- Arrow recipes
- GL36 algorithm stubs (doc-only)

# 57 JSON API driver

# 58 JSON API driver

Poll **HTTP or HTTPS** REST endpoints on the OT LAN (or public APIs such as OpenWeatherMap) and store extracted JSON values in the feather historian (`source=json_api`).

Use this driver for gateways, micro-PLCs, IoT hubs, vendor APIs, and **web weather** when you want to cross-check shaky local outdoor-air sensors against a reference.

## 58.1 Supported features

| Feature | Support |
|---|---|
| Methods | `GET`, `POST` |
| Transport | `http://` and `https://` |
| TLS certificate verify | On by default; disable for self-signed OT gateways |
| Authentication | None, **Bearer token**, **HTTP Basic**, or **query-string API keys via env** |
| Env placeholders | `${ENV:VAR}` or `${VAR}` in URL, headers, body, bearer token |
| Custom headers | Optional `headers` map (merged with auth headers) |

| Feature | Support |
|---|---|
| Value extraction | Dot-path into JSON body (`title`, `main.temp`, `weather.0.description`) |
| Multi-extract poll | One HTTP request per URL group; fan-out to multiple `json_path` rows |
| Polling | **1 / 5 / 15 / 30 min** or **1 hour** (shared with BACnet/Modbus) |
| Historian | `workspace/json_api/polls/samples.csv` → `feather_store/json_api/` |
| FDD merge | Scheduled FDD merges `bacnet` + `modbus` + `json_api` columns by nearest timestamp |

## 58.2 Secrets: `json_api.env.local`

Copy `workspace/json_api.env.example` → `workspace/json_api.env.local` (gitignored). The bridge loads this file at startup and when `run_local.sh` starts the stack.

Use placeholders in commissioning URLs so API keys never land in git:

```
https://api.openweathermap.org/data/2.5/weather?q=${ENV:OPENWEATHER_CITY}&appid=${ENV:OPENWEATHER_API_KEY}&units=${ENV:OPENWEATHER_UNITS}
```

Check which vars are configured: `GET /api/json-api/env/status`.

## 58.3 OpenWeatherMap showcase (recommended demo)

This is the flagship example of what the generic JSON API driver can do — **one URL, three historian columns**, env-based API key, browser tree like BACnet/Modbus.

### 58.3.1 1. Configure env

```
cp workspace/json_api.env.example workspace/json_api.env.local
# Edit: OPENWEATHER_API_KEY, OPENWEATHER_CITY, OPENWEATHER_UNITS (imperial = °F)
```

Restart the bridge (or `./scripts/run_local.sh restart`) so env vars load.

### 58.3.2 2. Register from the UI

1. Sign in as **integrator → JSON API** tab.
2. Open the **OpenWeatherMap showcase** panel.
3. Use the **OpenWeatherMap** preset → **Register** (default poll 30 min; choose interval in the form).

Three endpoints appear under `api.openweathermap.org` in the commissioning tree:

| Label | JSON path | Historian column | Units |
|---|---|---|---|
| web-oat-t | `main.temp` | `web-oat-t` | degF / degC |
| web-rh | `main.humidity` | `web-rh` | % |
| web-weather-desc | `weather.0.description` | `web-weather-desc` | text |

Poll worker deduplicates: **one GET** serves all three extractions per cycle.

### 58.3.3 3. Headless / script

```
        # Standalone fetch (like the original playground tester)
        python3 scripts/openweather_json_api_example.py

        # Register via REST (integrator login)
        python3 scripts/openweather_json_api_example.py --register --base
http://127.0.0.1:8000
```

### 58.3.4 4. FDD: local OA-T vs web weather

With BACnet `oa-t` and JSON API `web-oat-t` in the same site, scheduled FDD merges both sources. Example rule:

`workspace/data/rules_py/oat_vs_web_spread_1h.py` — flags when `|oa-t − web-oat-t| > 8 °F` (stuck or drifting outdoor-air sensor).

Enable it in Rule Lab, bind to your OA-T BACnet point, and run **Update all records**.

# 58.4 Commissioning (UI)

The **JSON API** tab follows Home Assistant `sensor.rest`:

1. Sign in → **RESTful sensor (JSON API)** tab.
2. Pick a **preset** (JSONPlaceholder, DummyJSON, httpbin, Open-Meteo, OpenAQ, OpenWeather) or enter a **resource URL**.
3. Add one or more **sensors** — each row is a `value_json_path` + historian **label** (multi-value from one GET, like HA `json_attributes`).
4. Click **Test resource** — probes without saving; shows extracted values and raw JSON.
5. Click **Register & poll** — writes `endpoints.csv`, enables polling at the selected interval.
6. Poll choices match BACnet/Modbus: **1 / 5 / 15 / 30 min** or **1 hour**.

## 58.4.1 API for agents

| Endpoint | Purpose |
|---|---|
| `GET /api/json-api/presets` | Full preset catalog + poll intervals (human + AI readable) |
| `POST /api/json-api/test` | Probe resource + multi-sensor extraction |
| `POST /api/json-api/register-bundle` | Register one resource with many sensors |
| `POST /api/json-api/presets/{id}/register` | One-click preset registration |

## 58.4.2 Bench preset

Use **JSONPlaceholder — todo title** → **Test resource** to verify outbound HTTPS without field hardware.

# 58.5 Authentication examples

## 58.5.1 Query-string API key via env (OpenWeather)

```
        {
          "url": "https://api.openweathermap.org/data/2.5/weather?q=$
{ENV:OPENWEATHER_CITY}&appid=${ENV:OPENWEATHER_API_KEY}&units=$
{ENV:OPENWEATHER_UNITS}",
          "method": "GET",
          "json_path": "main.temp",
          "label": "web-oat-t",
```

```
  "auth_type": "none"
}
```

### 58.5.2 Bearer token (API key)

```
{
  "url": "https://192.168.10.50/api/v1/status",
  "method": "GET",
  "json_path": "sensors.zone_temp",
  "label": "zone-temp",
  "auth_type": "bearer",
  "bearer_token": "${ENV:GATEWAY_API_KEY}",
  "verify_tls": false
}
```

### 58.5.3 HTTP Basic

```
{
  "url": "http://192.168.10.50/data.json",
  "method": "GET",
  "json_path": "value",
  "label": "sensor-value",
  "auth_type": "basic",
  "basic_user": "operator",
  "basic_password": "changeme"
}
```

### 58.5.4 POST with JSON body

```
{
  "url": "https://gateway.local/api/read",
  "method": "POST",
  "json_path": "result.pv",
  "label": "pv",
  "body": { "point": "AHU-1_SAT", "command": "read" },
  "auth_type": "bearer",
  "bearer_token": "…"
}
```

Credentials in `endpoints.csv` are redacted in API responses (***). Prefer `${ENV:…}` for production keys.

## 58.6 REST API

| Method | Path | Role | Description |
|---|---|---|---|
| GET | `/api/json-api/env/status` | operator+ | Which env vars are configured (no secret values) |

| Method | Path | Role | Description |
|---|---|---|---|
| POST | `/api/json-api/presets/openweather` | integrator+ | Register 3-point OpenWeather bundle + optional poll |
| GET | `/api/json-api/driver/tree` | operator+ | Commissioning tree grouped by host |
| GET | `/api/json-api/poll/status` | operator+ | Poll worker status |
| POST | `/api/json-api/poll/once` | integrator+ | Run one poll cycle |
| POST | `/api/json-api/request` | operator+ | One-shot HTTP request (no store) |
| POST | `/api/json-api/read_and_store` | integrator+ | Request + CSV + feather ingest |
| POST | `/api/json-api/refresh` | operator+ | Re-fetch one endpoint |
| PATCH | `/api/json-api/endpoint/poll` | integrator+ | Enable/disable poll interval |
| DELETE | `/api/json-api/endpoint/{point_id}` | integrator+ | Remove endpoint |

Full route list: API routes.

## 58.7 Typical OT URLs

| Device type | Example URL | Notes |
|---|---|---|
| Web weather | `https://api.openweathermap.org/data/2.5/weather?...` | Env-based `appid`; 3 historian columns |
| Gateway status | `http://192.168.1.50/api/status` | Often plain HTTP on LAN |
| HTTPS BMS API | `https://10.0.0.12/rest/points/zone1` | May need `verify_tls: false` |
| Local mock | `https://jsonplaceholder.typicode.com/todos/1` | Bench / CI only |

## 58.8 Limitations

- Response body must be **JSON** (not XML or plain text).
- Numeric paths (e.g. `main.temp`) are stored as strings in poll CSV; FDD rules cast with `pc.cast(..., "float64")`.
- Outbound requests originate from the **bridge** container/host — ensure routing and firewall rules allow the edge box to reach the target (OT subnet or public internet for weather).

## 58.9 Disable polling (save API quota)

Stop scheduled calls without deleting endpoints: JSON API tree → select OpenWeather points → **Stop polling**, or integrator API:

```
curl -X PATCH http://127.0.0.1:8765/api/json-api/endpoint/poll \
  -H "Authorization: Bearer $TOKEN" \
  -d '{"point_id":"ja-api-openweathermap-org-get-web-oat-t","enabled":false,"poll_interval_s":0}'
```

API keys live in `json_api.env.local` (gitignored) — never committed; audit logs do not record expanded URLs with `appid=`. See Logging and audit.

## 58.10 Smoke test

```
python3 scripts/smoke_multi_driver_stack.py
```

Validates JSON API ingest alongside BACnet and Modbus, feather isolation, BRICK scope, and FDD batch.

# 59 Algorithms

# 60 Algorithms (coming soon)

The **Algorithms** dashboard tab (`/algorithms`) is a placeholder for **supervisory Python sequences** — the mirror image of Rule Lab FDD:

| | FDD (Rule Lab) | Algorithms (planned) |
|---|---|---|
| Purpose | Detect faults | Compute setpoints, request levels, plant enables |
| Entrypoint | `apply_faults_arrow(table, cfg, context)` | `apply_algorithm_arrow(table, cfg, context)` (planned) |

| | FDD (Rule Lab) | Algorithms (planned) |
|---|---|---|
| Output | Boolean fault mask | Numeric outputs / structured dict |
| Historian | Same `feather_store` PyArrow tables | Same |

Until the tab ships, draft patterns live in this doc and in GL36 algorithm stubs.

# 60.1 GL36 Trim & Respond reference

Open-FDD algorithms will follow the same ASHRAE Guideline 36 trim & respond methodology already implemented for Niagara in:

**README_TRIM_RESPOND.md**

That guide covers:

- VAV zone request generators (cooling + static pressure)
- AHU duct static pressure reset
- AHU supply air temperature reset
- Central plant chiller / boiler enable and HWST / CHWST trim & respond

Porting those blocks to PyArrow is in progress; the dashboard tab shows **COMING SOON** until upload, batch, and BACnet write paths are wired.

# 60.2 Planned workflow (same kit shape as Rule Lab)

1. Download **algorithm kit** zip from the Algorithms tab (future)
2. Edit **constants** at the top of `algorithm.py` (no `config.json`)
3. Run `python run_test.py` locally against `sample.feather`
4. Upload `algorithm.py` when satisfied

Kit files will match the Rule Lab zip layout (see Rule Lab — dev kit zip).

# 60.3 Lookback windows

Rules and algorithms that need multi-hour history should use the shared **lookback stats helper** documented in Lookback window helper. Pass `hours=1`, `3`, `6`, `12`, or `24` to print row count, timestamp start/stop, and span for console validation.

Scheduled batch defaults: **1 h** lookback on the FDD loop; **Update all records** in Rule Lab uses **24 h** with 6 h chunks.

# 60.4 AHU supervisory checks (doc-only stubs)

These are **not** uploaded rules yet — copy-paste references for future algorithm or FDD work.

## 60.4.1 Duct static pressure too high

Flags sustained high duct static (possible stuck setpoint, blocked filter, or trim/respond not trimming).

```
        """AHU duct static high — doc stub (Arrow constants)."""

        import pyarrow.compute as pc

        VALUE_COLUMN = "duct-static"
        HIGH_INWC = 1.20
        LOOKBACK_HOURS = 3
        MIN_SAMPLES_ABOVE = 6


        def _kit_lookback_stats(table, *, hours=None):
            h = hours if hours is not None else LOOKBACK_HOURS
            ts = pc.cast(table["timestamp"], "timestamp[us, UTC]")
            tmin = pc.min(ts).as_py()
            tmax = pc.max(ts).as_py()
            span_h = (tmax - tmin).total_seconds() / 3600.0 if tmin and
tmax else 0.0
            print(f"lookback={h}h rows={table.num_rows} start={tmin}
stop={tmax} span={span_h:.2f}h")


        def apply_faults_arrow(table, cfg, context=None):
            _kit_lookback_stats(table)
            vals = pc.cast(table[VALUE_COLUMN], "float64")
            high = pc.greater(vals, HIGH_INWC)
            # Require several consecutive highs (~30 min at 5 min poll)
            from open_fdd.arrow_runtime.windows import
arrow_consecutive_true

            streak = arrow_consecutive_true(high, MIN_SAMPLES_ABOVE)
            return streak
```

### 60.4.2 Supply air temperature too cold (cooling coil over-delivering)

```
        """AHU SAT too cold vs setpoint — doc stub."""

        import pyarrow.compute as pc

        SAT_COLUMN = "sa-t"
        SP_COLUMN = "sa-t-sp"
        COLD_DELTA_F = 5.0
        LOOKBACK_HOURS = 1


        def _kit_lookback_stats(table, *, hours=None):
            h = hours if hours is not None else LOOKBACK_HOURS
            ts = pc.cast(table["timestamp"], "timestamp[us, UTC]")
            tmin = pc.min(ts).as_py()
            tmax = pc.max(ts).as_py()
            span_h = (tmax - tmin).total_seconds() / 3600.0 if tmin and
tmax else 0.0
            print(f"lookback={h}h rows={table.num_rows} start={tmin}
stop={tmax} span={span_h:.2f}h")


        def apply_faults_arrow(table, cfg, context=None):
            _kit_lookback_stats(table)
            sat = pc.cast(table[SAT_COLUMN], "float64")
            sp = pc.cast(table[SP_COLUMN], "float64")
            return pc.less(pc.subtract(sat, sp), -COLD_DELTA_F)
```

# 60.5 Plant supervisory checks (doc-only)

See GL36 algorithm stubs for chiller plant enable, HWST trim & respond, and CHW DP/CHWST reset patterns aligned with the Niagara README.

# 60.6 Related

- Rule Lab — live FDD authoring
- Model workflow — point bindings
- Windowing & debugging — rolling windows

# 61 Model workflow

# 62 Model workflow

## 62.1 Brick model

- **Equipment tree** — SPARQL over synced TTL (`workspace/data/data_model.ttl`)
- **Point registry** — BACnet inventory → `series_id` for historian bindings
- **Health score** — orphan points, missing `fdd_input`, stale telemetry

## 62.2 Typical integrator flow

1. Import or sync BACnet point registry.
2. Map equipment classes (AHU, VAV, …) in the model UI — set `brick_type` on points and `feeds` on equipment (AHU→VAV).
3. Tag points with `fdd_input` roles (zone temp, SAT, damper cmd, …).
4. Bind Rule Lab rules to points, equipment, or BRICK classes (`bindings` in `rules_store.json`).
5. Run `POST /api/model/sync-ttl` after bulk imports.
6. Validate on **Trend plot** with FDD overlays (`?fdd=1`) — faults render on a right-hand 0/1 axis per device scope.

AI Agent and commissioning JSON import can bulk-apply steps 2–4; integrator should review model health before production FDD. See FDD and assignments and Ollama and analytics.

Export: `GET /api/model/export` (TTL/JSON). Commissioning bundle: `GET /api/model/commissioning-export` — includes per-point `fdd_rules_linked` (Rule Lab names) plus `fdd_rules[]` catalog. Use **Import / export** tab preview table or **Point → FDD rule pins** before editing raw JSON. API detail in Appendix.

# 63 Driver framework

# 64 Driver framework

Open-FDD edge commissioning uses a **shared driver pattern** for OT data sources. Each driver polls field devices on a schedule, stores long-format samples in workspace CSVs, and ingests into the **feather historian** under a distinct `source` tag for FDD and plots.

## 64.1 Drivers

| Driver | Protocol | Feather `source` | Commissioning tab |
|---|---|---|---|
| BACnet | MS/TP / IP (: `47808`) | `bacnet` | BACnet |
| Modbus | Modbus TCP | `modbus` | Modbus |
| JSON API | HTTP/HTTPS REST (+ OpenWeather showcase) | `json_api` | JSON API |

All three drivers share the same operator workflow:

1. **Discover or configure** — add devices/endpoints to the driver registry.
2. **Request once** — on-demand read to verify connectivity and JSON path / register map.
3. **Request & store** — append to poll CSV and write a feather shard.
4. **Enable polling** — 1 / 5 / 10 / 15 minute intervals via tree right-click or bulk toolbar.
5. **Poll all now** — force one worker cycle without waiting for the interval.

## 64.2 Data layout

```
workspace/
  bacnet/commissioning/   points.csv, discovered inventory
  bacnet/polls/           samples.csv
  modbus/commissioning/   registers.csv
  modbus/polls/           samples.csv
  json_api/commissioning/ endpoints.csv
  json_api/polls/         samples.csv
```

```
data/feather_store/
  bacnet/{site_id}/     shard-*.feather
  modbus/{site_id}/
  json_api/{site_id}/
```

# 64.3 Historian and FDD

- **Plot tab** — numeric columns from `bacnet` and `modbus` sources (`GET /api/timeseries/plot?source=…`).
- **String JSON fields** — stored in feather under `json_api`; use the JSON API tree or feather directly (plot is numeric-only).
- **FDD batch** — merges `bacnet` + `modbus` + `json_api` historian columns by nearest timestamp; rules use BRICK-bound `external_id` names (e.g. compare BACnet `oa-t` vs JSON API `web-oat-t` — see JSON API).
- **BRICK model** — bind points in **Model & assignments** commissioning JSON; sync TTL for SPARQL scope.

# 64.4 Rule Lab (Arrow upload/download)

1. **Download kit** — zip with `rule.py` (constants at top), `data.py`, `sample.feather` (~3h), `run_test.py`, `requirements.txt`.
2. Local test: `pip install -r requirements.txt` then edit constants in `rule.py` and `python run_test.py`.
3. **Upload rule.py** — Arrow-only; must define `apply_faults_arrow(table, cfg, context=None)`.
4. Browser shows **read-only** source; use Quick test + Update all records.

API: `GET /api/rules/export-kit`, `POST /api/rules/upload`.

# 64.5 Smoke validation

From repo root on a running stack (`http://127.0.0.1:8765`):

```
python3 scripts/smoke_multi_driver_stack.py
python3 scripts/validate_modbus_temp_e2e.py --no-server
```

# 64.6 Related docs

- Data flow — poll → CSV → feather → FDD
- API routes — REST reference
- BACnet polling — commission container loop

# 65 Operator Bridge

# 66 Operator Bridge

The Operator Bridge is the FastAPI service and React dashboard operators use daily.

| Page | Topic |
|---|---|
| Dashboard overview | Main UI areas |
| Auth and roles | Login and permissions |
| Rule Lab | Python FDD rule authoring (PyArrow kit zip) |
| Algorithms | Supervisory GL36 sequences (coming soon) |
| Model workflow | Brick bindings and imports |
| FDD and assignments | Rule bindings, BRICK classes, commissioning JSON |
| Ollama and analytics | Home briefing, zone levers, model RAM tiers |

REST details: Appendix — API routes. Env files: Workspace env files.

# 67 Ollama and analytics

# 68 Ollama and analytics

## 68.1 What runs without login

The **Building status** home page (`/`) and `GET /openfdd-agent/building-insight` are **public** (same policy as fault status). Anyone on the LAN can read the briefing sentence and underlying numeric levers; only the **AI Agent** chat and stack revision endpoints require sign-in.

## 68.2 Ollama setup

1. Copy `workspace/ollama.env.example` → `workspace/ollama.env.local`.
2. Pick a model for available RAM (comments in the example file):

| RAM tier | Suggested model |
|---|---|
| 4 GB Pi | `qwen3:0.6b` |
| 8 GB | `qwen3:1.7b` |
| 16 GB | `qwen3:4b` |
| 32 GB | `qwen3:8b` |
| 64 GB | `qwen3:14b` |

1. Set `OFDD_OLLAMA_BASE_URL` (default `http://127.0.0.1:11434`).
2. Optional: `OFDD_OLLAMA_GPU_MODE` (`cpu` / `auto` / `gpu`), `OFDD_OLLAMA_TIMEOUT_S` for slow CPUs.
3. Bootstrap helper: `./scripts/bootstrap_ollama.sh`.

Bridge reads env via `run_local.sh` / compose. Model resolution: `workspace/api/openfdd_bridge/ollama_profiles.py`.

## 68.3 Automatic analytics (deterministic, always on)

These pandas calculations run on a refresh interval and feed the home dashboard **whether or not Ollama is up**:

| Lever | Module | Refresh env | What it compu |
|---|---|---|---|
| Zone temperature snapshot | `zone_temp_analytics.py` | `OFDD_ZONE_TEMP_INTERVAL_S` (default 3600s) | Site-wide and p AHU zone temp averages, sprea out-of-band cou using BRICK `fe` (AHU→VAV) wh modeled; falls b to all `Zone_Air_Tempera` points |
| Device poll health | `device_poll_health.py` | stack health cycle | BACnet device l seen / stale poll detection |
| Operational brief | `building_insight.py` → `get_operational_brief` | same as insight | Compact numer status for integrators |
| | `root_cause_hints.py` | on insight refresh | |

| Lever | Module | Refresh env | What it compu |
|---|---|---|---|
| Root-cause hints | | | Multi-zone fault chains using `brick:feeds` from |

Shared lookback window: `OFDD_ANALYTICS_LOOKBACK_DAYS` (default 14) in `operational_analytics.py`.

API routes (auth optional unless noted):

- `GET /openfdd-agent/building-insight` — sentence + zone snapshot + fault sentences
- `GET /openfdd-agent/zone-temp-analytics`
- `GET /openfdd-agent/device-poll-health`
- `GET /openfdd-agent/ollama/health` — integrator diagnostics

# 68.4 When Ollama is used

`building_insight.get_building_insight()`:

1. Refreshes zone + device snapshots (pandas).
2. Calls `ollama_client.health()`.
3. If Ollama is reachable and `should_use_ollama_for_insight()` passes, sends a **compact context** (status, zone levers, feeds chains) to the configured model for a one-sentence briefing (`source: "ollama"`).
4. On failure, uses `_fallback_sentence()` (`source: "deterministic"`).

Context assembly: `building_insight._compact_context()`. Prompt template references BRICK feeds and `root_cause_hints`.

# 68.5 AI Agent (authenticated)

`/agent` page uses `POST /openfdd-agent/chat` with tool access (`agent_tools.py`): BRICK graph, scope bundles, rule binding patches, model health, etc. This is separate from the public home briefing.

# 68.6 Code map

```
workspace/api/openfdd_bridge/
  building_insight.py      # cached home briefing + Ollama
sentence
  zone_temp_analytics.py   # zone temp levers (BRICK feeds)
  device_poll_health.py    # BACnet poll staleness
  operational_analytics.py # lookback window + methodology dict
  root_cause_hints.py      # multi-fault BRICK chain hints
```

```
  ollama_client.py         # HTTP client to Ollama
  ollama_profiles.py       # RAM tiers / model names
```

Dashboard consumers: `BuildingStatusPage.tsx`, `StackStatusStrip.tsx` (WebSocket `/ws/dashboard`).

# 69 FDD and assignments

# 70 FDD and assignments

## 70.1 What the product actually does

Open-FDD separates **data modeling** (BRICK equipment, points, `feeds` relationships) from **rule assignment** (which FDD rules apply to which points, equipment, or BRICK classes). Both are editable in **Model & assignments** and via commissioning JSON import.

### 70.1.1 BRICK modeling (integrator + AI-assisted)

| Concern | Where it lives | Notes |
|---|---|---|
| Equipment tree | `ModelService` + TTL sync | AHU, VAV, sensors; `equipment_type` and `brick_type` on points |
| `brick:feeds` | `model_feeds.py`, `ttl_service.py` | AHU→VAV→zone relationships; used by zone analytics and root-cause hints |
| Point registry | BACnet inventory → model points | `fdd_input` roles (`oa-t`, `stat_zn-t`, …), `series_id` / historian column |
| Commissioning import | `POST /api/model/commissioning-import` | Bulk JSON: sites, equipment, points, optional `fdd_rules` |

**AI Agent** (`agent_tools.py`) can read `slim_brick_graph`, `scope_bundle`, patch rule bindings, and upsert points — but it does **not** auto-generate a full site model without operator review. Expected workflow:

1. BACnet discover/poll populates point registry.
2. Integrator or agent tags `brick_type`, `fdd_input`, and `feeds` on equipment.
3. `POST /api/model/sync-ttl` writes `workspace/data/data_model.ttl`.
4. Model health (`model_health.py`) flags orphan points and missing `fdd_input`.

## 70.1.2 Rule assignment (three binding kinds)

Saved rules in `rules_store.json` carry `bindings`:

```
{
  "point_ids": ["5007-analog-input-1173"],
  "equipment_ids": ["bench-1"],
  "brick_types": ["Outside_Air_Temperature_Sensor"]
}
```

| Binding kind | Effect |
|---|---|
| `point_ids` | Rule runs for that historian point / series |
| `equipment_ids` | Rule runs for all points on equipment (mass bind) |
| `brick_types` | Rule runs for any point with that BRICK class on site |

Assign via:

- **Model & assignments** UI — equipment/point chips, rule pin menu on trend plot
- **Rule Lab** — bindings panel
- **Commissioning import** — `fdd_rules[].bindings` and per-point `fdd_rule_ids`
- **AI Agent** — `patch_rule_binding` tool

Runtime evaluation: `fdd_runner.py` (batch FDD) and `plot_readings.evaluate_fault_plots()` (trend overlays). Plot scope filters rules to those bound to **selected telemetry** (point, equipment, or BRICK class); unbound rules still evaluate site-wide.

### 70.1.3 CLASS vs device

- **BRICK class** (`brick_type` on points, or `brick_types` on rule bindings) — template-style assignment ("all `Zone_Air_Temperature_Sensor`").
- **Device / equipment** (`equipment_id`, BACnet device instance) — instance assignment ("this VAV-12").
- **Point** (`point_ids`) — single-sensor assignment.

`applies_to` on legacy rules is deprecated; use `bindings` in 3.0+.

## 70.2 Commissioning JSON shape

Export: `GET /api/model/commissioning-export`. Import accepts `import_ready_json` wrappers (see `commissioningImport.ts`).

### 70.2.1 Rule names vs ids (Rule Lab alignment)

Rule Lab shows **human names** (`Bench OA-T flatline 1h`). Saved rules use stable **ids** (`bench-oa-t-flatline-1h` from setup scripts, or a UUID for uploaded custom rules). Commissioning export includes both so you never have to cross-reference sections by hand:

| Field | Direction | Purpose |
|---|---|---|
| `points[].fdd_rule_ids` | import + export | Assignment surface — array of rule ids |
| `points[].fdd_rules_linked` | **export only** | `[{id, name}]` — Rule Lab names beside each point |
| `fdd_rules[].id` | import + export | Stable rule key (matches Rule Lab / `rules_store.json`) |
| `fdd_rules[].name` | import + export | Same label as Rule Lab dropdown |
| `fdd_rules[].source_file` | export | Basename of deployed `.py` when present (e.g. `bench_oa-t_flatline_1h.py`) |

On import, `fdd_rules_linked` is stripped (like `fdd_rule_ids` — not stored in `model.json`). Use the **Point → FDD rule pins** table on Model & assignments to read names without parsing JSON.

Minimum fields integrators/AI should produce:

- `sites`, `equipment[]` with `id`, `equipment_type`, optional `feeds[]`
- `points[]` with `site_id`, `equipment_id`, `brick_type`, `fdd_input`, BACnet IDs, optional `fdd_rule_ids`
- `fdd_rules[]` with `id`, `name`, `bindings`, `enabled` (optional `source_file` on export)

After import, verify in **Trend plot** with `?fdd=1` — fault lanes appear on the right-hand 0/1 axis for bound rules.

## 70.3 Related

- Model workflow
- Rule Lab
- Trend plot / fault overlays

# 71 Network setup

# 72 Network setup

## 72.1 Bind address

Commission and poll containers need the **OT interface** IP and UDP **47808**:

```
# workspace/bacnet/commissioning/commission.env
BACNET_BIND=192.168.1.50/24:47808
BACNET_INSTANCE=599999
```

- Use the NIC that can reach controllers (not only the management VLAN unless routed).
- Poll driver uses `network_mode: host` on Linux edges.

## 72.2 Discovery range

```
DISCOVER_LOW=1
DISCOVER_HIGH=4194302
DISCOVER_TIMEOUT=30
```

Narrow the range in large sites to avoid broadcast storms.

## 72.3 Verify

From commission container logs or API: Who-Is → I-Am responses. If zero devices, check firewall, bind IP, and VLAN routing before tuning rules.

# 73 Discover and read

# 74 Discover and read

The **commission** container exposes HTTP APIs for discovery and read-property workflows used by the dashboard BACnet tools.

| Capability | Notes |
|---|---|
| Who-Is / I-Am | Device discovery on OT subnet |
| Read property | Present value, object name, units |
| RPM | Bulk reads for inventory export |
| Write property | **Gated** — see Write safety |

Exported inventory feeds `points_discovered.csv` and the Brick point registry.

## 74.1 Operator workflow

1. Confirm network setup.
2. Run discover from dashboard or API.
3. Review discovered objects; enable points for poll profiles.
4. Sync model bindings for FDD inputs.

# 75 Polling

# 76 Polling

**BACnet field reads run in the `openfdd-commission` container.** The Operator Bridge does not bind UDP 47808.

> **Retired:** `openfdd-bacnet-poll` image and `openfdd-bacnet-poll` systemd unit. Poll loop lives in **commission** only — do not run both.

BACnet polling has two stages:

1. **Field reads** — commission poll loop → `workspace/bacnet/polls/samples.csv`
2. **Historian ingest** — bridge background worker → `workspace/data/feather_store/`

# 76.1 Commission poll loop (default)

The **commission** container runs the poll loop at startup. It reads enabled rows from `points.csv` and appends BACnet RPM results to `samples.csv`.

| Setting | Location |
|---|---|
| BACnet bind | `workspace/bacnet/commissioning/commission.env` |
| Enabled points | `workspace/bacnet/commissioning/points.csv` |
| Poll output | `workspace/bacnet/polls/samples.csv` |
| Historian | `workspace/data/feather_store/` |

Check activity:

```
tail -1 workspace/bacnet/polls/samples.csv
docker compose logs --since 5m commission | grep -i poll
```

# 76.2 Bridge ingest worker

Inside **bridge**, a background thread watches `samples.csv` and ingests new rows into feather. Disable only for debugging: `OFDD_DISABLE_POLL_WORKER=1`.

# 76.3 Health

Dashboard stack health (`GET /health/stack`) reports `bacnet_poll` status from the commission agent — not a separate container.

Stale points trigger data-quality fault patterns — see Sensor quality faults.

# 77 Write safety

# 78 Write safety

> **BACnet writes can change live plant behavior.** OpenFDD disables writes by default. Enable only with operator sign-off, a tested allowlist, and audit logging.

## 78.1 Defaults

| Control | Default |
|---|---|
| Supervisory writes | **Off** |
| Write allowlist | Empty — must enumerate device/object |
| Audit | Commands logged when writes enabled |

## 78.2 Enable writes (commission role)

1. Set env flag documented in Security (`OFDD_BACNET_WRITE_ENABLED` or site equivalent).
2. Populate write allowlist CSV / config with explicit object IDs.
3. Use **commission** role JWT; every write should include reason/note where API supports it.
4. Test on bench hardware before production OT.

## 78.3 Operator rules

- Never write setpoints or overrides on life-safety or fire/smoke interlocks through experimental rules.
- Prefer read-only polling for FDD until rules are validated in Rule Lab.
- Revert overrides through BACnet priority relinquish where supported.

# 79 BACnet

# 80 BACnet

Open-FDD integrates with field controllers via BACnet/IP (and bench MSTP overlays in dev compose).

| Page | Topic |
|---|---|
| Network setup | Bind address and NIC |
| Discover and read | Commission agent |
| Polling | Historian driver |
| Write safety | Defaults and allowlists |

# 81 Convention

# 82 Fault code convention

## 82.1 Format (Grade-A, 3.0.18+)

**Short stable code:** `FAMILY-SUBSYSTEM-NNN` — e.g. `AHU-ECON-001`, `CHW-DT-001`, `DATA-STAL-001`.

**Semantic ID:** dotted `canonical_id` — e.g. `ahu.economizer.not_using_free_cooling`.

**Legacy letter codes** (`AHU-E`, `VAV-C`) remain **aliases** in `open_fdd/faults/catalog/*.yaml` until bridge sync completes.

| Field | Example |
|---|---|
| `code` | `AHU-ECON-001` |
| `canonical_id` | `ahu.economizer.not_using_free_cooling` |
| `family` | `AHU`, `VAV`, `CHW`, `DATA`, `CRS`, … |
| `rule_doc_path` | Link to rule cookbook recipe |

Full schema: `open_fdd/faults/schema.py`. Catalog loader: `open_fdd/faults/catalog.py`.

## 82.2 Categories (Grade-A)

| Category | Meaning |
|---|---|
| `energy` | Waste, reset stuck, plant inefficiency |
| `comfort` | Setpoint not met, poor zone performance |
| `reliability` | Short cycling, staging, equipment stress |
| `maintenance` | Leaking valves, filters, fouling proxies |
| `indoor_air_quality` | Ventilation shortfall, humidity |
| `healthcare_risk` | Pressure, isolation, critical-space excursions |
| `data_quality` | Stale, flatline, OOB, missing historian |
| `controls_integrity` | Overrides, chatter, schedule violations |

## 82.3 Severity

| Level | Operator response |
|---|---|
| `info` | Log; trend for maintenance window |
| `low` | Review in routine rounds |
| `medium` | Investigate within days |
| `high` | Prompt attention; comfort/energy impact likely |
| `critical` | Immediate operational risk (especially `healthcare_risk`) |

## 82.4 Linking rules

In Rule Lab, set `fault_code` on the rule metadata. Batch FDD aggregates hits into `GET /api/faults/status` by family.

## 82.5 Cookbook mapping

Each catalog entry lists `cookbook_patterns` (e.g. `flatline_1h`, `mixing_envelope`, `rate_of_change`) — see Expression cookbook (Arrow-native).

Legacy pandas `type: expression` YAML rules are **not** used on the edge in 3.x; translate to `rules_py` Arrow modules and bind **`fault_code`** in Rule Lab.

# 83 Live site update

# 84 Live site update

Upgrade GHCR images on a **live edge host** that already has `~/open-fdd/` — no `git pull` on the host.

> **IT operators:** prefer Quick Start — Updating the stack and `openfdd_site_backup.sh` / `openfdd_site_update.sh` on the edge host. This page adds control-machine UI deploy and Ansible paths.

## 84.1 Layout on the edge host

```
~/open-fdd/
  docker-compose.yml
  workspace/          # site state — backup first
```

`workspace/` holds BACnet commissioning, poll CSV, feather historian, BRICK model, FDD rules, and `auth.env.local`. Image upgrades must **preserve** it.

See Backup and restore before any upgrade.

## 84.2 Concepts

| Term | Meaning |
|---|---|
| **Container** | Running instance (`bridge`, `commission`, `mcp-rag`) — recreated on upgrade |
| **Image** | GHCR package (tag `latest` or dated tag) |
| **`workspace/`** | Open-FDD site state on disk — **do not delete** |

## 84.3 1. Verify new images on GHCR

Three images must exist for every tag:

```
        export NEW_TAG="${NEW_TAG:-latest}"

        docker manifest inspect ghcr.io/bbartling/openfdd-bridge:$
{NEW_TAG} >/dev/null &&
        docker manifest inspect ghcr.io/bbartling/openfdd-commission:$
```

```
{NEW_TAG} >/dev/null &&
        docker manifest inspect ghcr.io/bbartling/openfdd-mcp-rag:$
{NEW_TAG} >/dev/null &&
        echo "All three images exist for ${NEW_TAG}"
```

## 84.4 2. Update dashboard UI (when release changed React)

> **Image pull alone does not update the browser UI.** The Operator Bridge serves `workspace/api/static/app/` from the bind mount **before** image-baked assets.

**From control machine (full repo):**

```
cd /path/to/open-fdd
./scripts/build_operator_dashboard.sh prod
cd infra/ansible
RUN_POST_CHECK=0 ./deploy.sh ui --limit <inventory_host>
```

Confirm hash on edge:

```
curl -sf http://<edge-ip>/ | grep -o 'index-[^"]*\.js' | head -1
```

**One command (UI + images + insurance check):**

```
OPENFDD_IMAGE_TAG=latest ./scripts/upgrade_edge_full.sh --limit
<inventory_host>
```

## 84.5 3. Pull and recreate (edge host)

```
cd ~/open-fdd
./scripts/openfdd_site_backup.sh
./scripts/openfdd_site_update.sh
```

Manual equivalent:

```
cd ~/open-fdd
docker compose pull
docker compose up -d --force-recreate
docker compose ps
curl -sf http://127.0.0.1:8765/health
```

## 84.6 4. Ansible image-only (control machine)

```
OPENFDD_IMAGE_TAG=latest RUN_POST_CHECK=1 ./scripts/
upgrade_edge_ghcr.sh --limit <inventory_host>
```

## 84.7 5. Validate

→ Deployment validation

## 84.8 Rollback

Restore `docker-compose.yml` from backup, pull previous tag, `docker compose up -d --force-recreate`. `workspace/` is unchanged. Details: Backup and restore.

## 84.9 Related

- Updating the stack — edge-host helper scripts
- Containers — three-image architecture

# 85 AHU faults

# 86 AHU faults

Air handling unit codes (sample from catalog).

| Code | Title | Severity | Detection summary | Likely causes | Operator checks |
|---|---|---|---|---|---|
| **AHU-A** | Supply fan performance degradation | warning | Spread/ runtime vs baseline | Belt slip, dirty filters, VFD limit | Trend speed vs airflow; filter dP |
| **AHU-B** | Simultaneous heating and cooling | critical | Heat + cool outputs active together | Sequencing bug, leaking valve | Trend coil commands; valve feedback |
| **AHU-C** | SAT sensor fault | warning | Flatline / OOB SAT | Failed sensor, bad calibration | SAT trend; compare to coil discharge |
| **AHU-D** | MAT inconsistent with OAT/ RAT | warning | MAT outside OAT–RAT envelope | MAT/OAT/ RAT sensor errors | Verify damper positions; cross-check OAT |

| Code | Title | Severity | Detection summary | Likely causes | Operator checks |
|---|---|---|---|---|---|
| **AHU-E** | Economizer not economizing | warning | Free cool available, OA minimum | Stuck damper, lockout | OA damper vs OAT trend |
| **AHU-F** | Damper command vs feedback mismatch | warning | Cmd ≠ feedback | Stuck actuator, linkage | Compare cmd/ feedback trends |

**Example rule:** `flatline_1h` on SAT → tag **AHU-C**. Recipe: Python recipes.

# 87 Standards crosswalk

# 88 Standards crosswalk

Grade-A fault definitions carry a `standards_crosswalk` block in `open_fdd/faults/catalog/*.yaml`.

| Field | Source | Use in Open-FDD |
|---|---|---|
| `ornl_lbnl_taxonomy` | ORNL/ LBNL unified HVAC fault taxonomy | Canonical ontology backbone |
| `ashrae_g36_reference_type` | ASHRAE Guideline 36 | Trim-and-respond, plant requests, supervisory sequences |
| `ashrae_207_reference_type` | ASHRAE Standard 207 | Economizer FDD test categories |
| `brick_classes` | Brick Schema | Equipment/point semantic roles |
| `haystack_tags` | Project Haystack (optional) | Interoperability tags |

## 88.1 Example

```
standards_crosswalk:
  ornl_lbnl_taxonomy: ahu/economizer/free_cooling
  ashrae_g36_reference_type: supply_air_temp_reset
  ashrae_207_reference_type: economizer_fault_detection
  brick_classes: [brick:Air_Handler_Unit, brick:Economizer]
  haystack_tags: [equip, ahu, econ]
```

Export full catalog: `python3 -c "from open_fdd.faults.catalog import catalog_export; import json; print(json.dumps(catalog_export(), indent=2))"`

# 89 Backup and restore

# 90 Backup and restore

`workspace/` on the edge host is the **live site state** — feather historian, BACnet commissioning files, BRICK model, FDD rules, auth, and dashboard static bundle. Container images are replaceable; **backup `workspace/` before every upgrade.**

## 90.1 Quick backup (edge host)

```
cd ~/open-fdd
./scripts/openfdd_site_backup.sh
```

Archives land under `~/openfdd-backups/<timestamp>/`.

Custom location:

```
BACKUP_ROOT=~/openfdd-backups/manual ./scripts/openfdd_site_backup.sh
```

## 90.2 Manual backup (SSH)

Container processes may write files as **`root:root`** with mode **`0600`**. Use `sudo` for a full archive:

```
cd ~/open-fdd

export BACKUP_ROOT="$HOME/openfdd-backups/$(date +%Y%m%d-%H%M%S)"
mkdir -p "$BACKUP_ROOT"
```

```
        cp docker-compose.yml "$BACKUP_ROOT/docker-compose.yml.before"
        docker compose ps > "$BACKUP_ROOT/docker-compose-ps-before.txt"
        docker compose config --images > "$BACKUP_ROOT/docker-images-
before.txt"

        sudo tar --xattrs --acls -czf "$BACKUP_ROOT/workspace-full.tgz"
workspace
        sudo chown "$USER:$USER" "$BACKUP_ROOT/workspace-full.tgz"

        du -h "$BACKUP_ROOT/workspace-full.tgz"
```

## 90.3 What to protect

| Path | Contents |
|---|---|
| `workspace/data/feather_store/` | Feather historian |
| `workspace/bacnet/commissioning/` | `points.csv`, `commission.env` |
| `workspace/bacnet/polls/samples.csv` | Poll output |
| `workspace/data/*.json` | Model, rules store, FDD results |
| `workspace/auth.env.local` | Login secrets |
| `workspace/api/static/app/` | Dashboard bundle (if deployed separately) |

## 90.4 Restore after a bad upgrade

1. Stop containers: `docker compose stop` (do **not** use `down -v`)
2. Restore `docker-compose.yml` from backup
3. If data corruption: extract selective files from `workspace-full.tgz` — model, rules, feather shards
4. `docker compose pull && docker compose up -d --force-recreate`
5. Verify: Deployment validation

## 90.5 Safe cleanup

```
        docker image prune -f
```

> **Never on live sites:** `docker compose down -v`, `docker volume prune`, `docker system prune -a --volumes`, or deleting `workspace/`.

## 90.6 Related

- Updating the stack
- Live site update

# 91 VAV faults

# 92 VAV faults

Variable air volume terminal codes (sample).

| Code | Title | Severity | Detection summary | Likely causes | Operator checks |
|---|---|---|---|---|---|
| **VAV-A** | Zone heating/ cooling overlap | warning | Heat and cool at terminal | Reheat valve stuck, SAT too low | Zone temp vs SAT; valve cmd |
| **VAV-B** | Poor zone temperature control | warning | Persistent deviation from setpoint | Oversized box, bad tuning | PI loop trends; occupancy |
| **VAV-C** | Zone temperature sensor fault | warning | Flatline / OOB zone temp | Failed zone sensor | Flatline test; compare to neighbor zones |
| **VAV-D** | Minimum airflow not met | warning | Flow below minimum cmd | Damper stuck, pressure issue | Airflow vs damper cmd |
| **VAV-E** | Discharge temp deviation from SAT | warning | DAT far from SAT setpoint | Coil issue, sensor drift | DAT vs SAT SP |

**Example rule:** `oob_rolling` on zone temp → **VAV-C** or comfort bounds per site policy.

# 93 RTU faults

# 94 RTU faults

Packaged rooftop unit codes (sample).

| Code | Title | Severity | Detection summary | Likely causes | Operator checks |
|---|---|---|---|---|---|
| **RTU-A** | Compressor short cycling | warning | Rapid on/off cycles | Low charge, oversized unit | Run-time histogram |
| **RTU-B** | Economizer / OA damper fault | warning | OA behavior vs conditions | Actuator, sensor | OA damper vs enthalpy/ OAT |
| **RTU-C** | Supply air temperature fault | warning | SAT flatline or OOB | Sensor failure | SAT trend vs outdoor conditions |
| **RTU-D** | Simultaneous heat and cool | critical | Heat + cool stages together | Control board, relay stuck | Stage status trends |

Bind rules to packaged unit `fdd_input` points (SAT, OAT, compressor cmd, …).

# 95 Deployment validation

# 96 Deployment validation

Non-destructive checks after deploy or image upgrade. Run from the **edge host** (smoke) or from a **control machine** with the full repo (insurance suite).

# 96.1 Edge host smoke (5 minutes)

```
cd ~/open-fdd

docker compose ps
curl -sf http://127.0.0.1:8765/health | jq . || curl -sf http://127.0.0.1:8765/health

tail -1 workspace/bacnet/polls/samples.csv
du -sh workspace/data/feather_store/

docker compose logs --since 20m | grep -Ei "error|exception|traceback|critical" | tail -20 || true
```

Via Caddy (building LAN):

```
curl -sf http://<edge-ip>/health
curl -sf http://<edge-ip>/ | grep -o 'index-[^"]*\.js' | head -1
```

# 96.2 Browser checklist

1. Open `http://<edge-ip>/` (Caddy `:80`, not only loopback `:8765`)
2. Integrator login (`workspace/auth.env.local` on the edge host)
3. **Host stats** — container image tag / git SHA
4. **BACnet** — driver tree shows devices; last poll timestamp recent
5. **Model & assignments** — commissioning export returns JSON
6. **Rule Lab** — quick-test a saved rule (`backend: arrow`)

# 96.3 Control-machine insurance check

From a machine with the full `open-fdd` repo and Ansible inventory:

```
cd infra/ansible/scripts
./post_deploy_check.sh --limit <inventory_host> --full
```

Or by IP:

```
./post_deploy_check.sh --host <edge-ip> --ssh-user <deploy-user> --full
```

Checks include: Caddy → React dashboard, `/health/stack`, MCP RAG, BACnet tree, BRICK/SPARQL, Docker log scan.

> **Do not run on live commissioned sites without approval:** `acme_operational_verify.sh` with BACnet rediscover — use `--skip-discover` for smoke only. See Examples — GL36 lab.

## 96.4 LAN security smoke (optional)

From a workstation on the same LAN as the edge:

- Windows: `scripts/security/Run-OpenFddSecurityScan.ps1`
- macOS/Linux: `scripts/security/run_openfdd_security_scan.sh`

Expected findings: ZAP baseline.

## 96.5 Arrow / FDD rules (3.0+)

On control machine:

```
./scripts/validate_fdd_backends.sh --docker
```

On edge after upgrade:

```
grep -R "apply_faults_arrow" workspace/data/rules_py | wc -l
```

## 96.6 Related

- Health check
- Security testing cycle (maintainers)

# 97 Sensor and data quality

# 98 Sensor and data quality

Sensor faults use cookbook patterns **`flatline_1h`**, **`oob_rolling`**, **`rate_of_change`**, and **`mixing_envelope`**. Full thresholds: Expression cookbook — sensor validation.

## 98.1 Catalog codes

| Code | Title | Severity | Pattern | Detection summary |
|---|---|---|---|---|
| **BLD-B** | Outdoor air sensor fault | warning | flatline, oob, rate_of_change | OAT flatline, OOB, or unrealistic spike |
| | | critical | stale_points | No new samples |

| Code | Title | Severity | Pattern | Detection summary |
|---|---|---|---|---|
| **BLD-D** | Stale telemetry | | | within threshold |
| **AHU-C** | SAT sensor fault | warning | flatline_1h, oob_rolling | Supply air temp stuck or out of band |
| **AHU-D** | MAT inconsistent with OAT/RAT | warning | mixing_envelope | Mixed air outside OAT–RAT envelope |
| **VAV-C** | Zone temperature sensor fault | warning | flatline_1h, oob_rolling | Zone temp stuck or 55–90 °F band breach |
| **RTU-C** | Supply air temperature fault | warning | flatline_1h, oob_rolling | Packaged unit SAT fault |
| **CH-D** | CHW supply temperature sensor fault | warning | flatline_1h, oob_rolling | Plant CHW sensor fault |
| **DC-C** | Datacom sensor anomaly | warning | flatline_1h | CRAH/room sensor flatline or RH OOB |
| **HP-D** | Discharge/suction temperature sensor fault | warning | flatline_1h, oob_rolling | Heat pump refrigerant/air temp sensor |

## 98.2 Default bounds (imperial, occupied)

| Sensor | Min | Max | Flatline tol | Max Δ/hr |
|---|---|---|---|---|
| Zone temp | 55 °F | 90 °F | 0.10 °F | 4 °F |
| Supply air temp | 50 °F | 110 °F | 0.15 °F | 8 °F |
| Return air temp | 55 °F | 95 °F | 0.10 °F | 3 °F |
| Duct static | −0.5 | 3.0 in$H_2O$ | 0.02 | 0.5 |
| RH | 0 % | 100 % | 1.0 % | 15 % |
| CHW | 40 °F | 90 °F | 0.10 °F | 4 °F |
| HW | 70 °F | 200 °F | 0.15 °F | 6 °F |
| | 50 °F | 110 °F | 0.15 °F | 5 °F |

| Sensor | Min | Max | Flatline tol | Max Δ/ hr |
|---|---|---|---|---|
| Condenser water | | | | |
| $CO_2$ (occupied) | 400 ppm | 1000 ppm | 5 ppm | 200 ppm |

Return-air checks use **narrow bands**; when MAT/OAT/RAT are available, prefer **mixing envelope** over RAT bounds alone.

## 98.3 Communication quality

- Monitor poll cycle timestamps in dashboard.
- Use `stale_points` for dropout detection (**BLD-D**).
- Do not confuse **comm loss** with **equipment fault** — verify BACnet device online before dispatching mechanical.

## 98.4 Operator workflow

1. Confirm **BLD-D** (stale) is resolved before trusting other sensor codes.
2. Cross-check **BLD-B** OAT against weather service or JSON API driver.
3. For **VAV-C** / **AHU-C**, compare neighbor zones / coil behavior before replacing sensor.

# 99 Logging and audit

# 100 Logging and audit

Open-FDD uses a **two-tier logging model** familiar to IT and security teams: structured **audit/error JSONL** for forensics and compliance, plus **service stdout** for engineering troubleshooting. Defaults include **age pruning**, **size rotation in MB**, and **Docker log caps** — no operator cron required.

## 100.1 Log types

| Tier | Location | Format | Who reads it |
|---|---|---|---|
| **Audit** | `workspace/logs/audit.jsonl` | JSON Lines | Security, MSI, integrator API |

| Tier | Location | Format | Who reads it |
|---|---|---|---|
| **Errors** | `workspace/logs/error.jsonl` | JSON Lines | Support, integrator API |
| **Service stdout** | Docker: `docker compose logs` · Dev: `workspace/.local-run/*.log` | Plain text (uvicorn, workers) | Engineering |
| **Historian** | `workspace/data/feather_store/` | Feather | FDD, plots — separate retention (`data.env.local`) |

Structured audit logs are **not** mixed into uvicorn access logs. That separation matches common practice: SIEM-friendly JSONL for auth and OT commands; unstructured stdout for stack traces and poll worker noise.

# 100.2 Audit record schema

Each line in `audit.jsonl` / `error.jsonl` is one JSON object:

| Field | Example | Notes |
|---|---|---|
| `@timestamp` | `2026-06-07T19:57:21+00:00` | UTC ISO-8601 |
| `event_id` | UUID | Unique per event |
| `event_type` | `auth.login.failure` | Stable taxonomy (see below) |
| `severity` | `warning` | `debug` … `critical` |
| `outcome` | `failure` | `success`, `failure`, `denied`, … |
| `service` | `openfdd-bridge` | |
| `host` | hostname | |
| `action` | `login` | Human-readable verb |
| `actor` | `{username, role}` | After successful auth |
| `client` | `{ip, user_agent}` | Respects `OFDD_TRUST_X_FORWARDED_FOR` behind Caddy |
| `http` | method, path, status | When HTTP-triggered |
| `resource` | BACnet object, model id, … | When applicable |
| `detail` | Sanitized map | **Passwords, tokens, secrets stripped by key name** |
| `request_id` | UUID | On audited HTTP paths |

### 100.2.1 Auth events (industry-standard login trail)

| event_type | When |
|---|---|
| auth.login.success | Valid credentials |
| auth.login.failure | Bad password (generic client message) |
| auth.login.lockout | Rate limit exceeded (default 5 failures / 5 min) |

Login rate limits and lockouts are **audited**; responses never reveal whether the username exists.

### 100.2.2 OT / commissioning events

| event_type | When |
|---|---|
| bacnet.command | Supervisory write (allowed, denied, dry-run) |
| bacnet.discover | Who-Is / point discovery |
| model.write | BRICK model mutations |
| agent.tool | Agent tool calls (prompts hashed/truncated) |
| api.access | Sensitive POST/PUT/PATCH/DELETE (BACnet, ingest) |
| error.unhandled | Unhandled exception (also in error.jsonl) |

**Not audited:** routine GETs (dashboard reads, plot data, health). This reduces noise while keeping **auth and field commands** on the trail.

### 100.2.3 Secret redaction

Never logged: passwords, bearer tokens, OFDD_AUTH_SECRET, private keys. Detail objects drop keys containing password, token, secret, authorization. Agent tool arguments use additional hashing for prompts and rule source.

Opt-in full prompt logging (lab only): OFDD_AUDIT_LOG_PROMPTS=1.

## 100.3 Integrator API (read logs in the UI stack)

| Method | Path | Role |
|---|---|---|
| GET | /api/audit/events?limit=100 | integrator |
| GET | /api/audit/errors?limit=100 | integrator |

| Method | Path | Role |
|---|---|---|
| GET | `/api/audit/summary` | integrator |

Use these for post-incident review without shell access. Forward `workspace/logs/*.jsonl` to SIEM (rsyslog, Filebeat, Wazuh) on production edges.

## 100.4 Retention and rotation (defaults)

Configured in `workspace/data.env.local` (copy from `data.env.example`):

| Variable | Default | Behavior |
|---|---|---|
| `OFDD_LOG_RETENTION_DAYS` | **90** | Drop audit/error JSONL lines older than N days; delete aged `.local-run` and archived `*.log` / `audit.*.jsonl` |
| `OFDD_LOG_MAX_MB` | **50** | When `audit.jsonl` or `error.jsonl` exceeds N MB, archive to `audit.{UTC}.jsonl` and start fresh |
| `OFDD_LOCAL_RUN_LOG_MAX_MB` | **25** | Rotate `workspace/.local-run/*.log` (bridge, Caddy, FDD loop, …) |
| `OFDD_LOG_ROTATE_INTERVAL_HOURS` | **6** | Background retention while bridge is running (not only at restart) |

Rotation runs at **bridge startup** and on the **scheduled interval**. No logrotate.d entry is required for audit JSONL.

Optional path overrides:

```
OFDD_AUDIT_LOG_PATH=/var/log/openfdd/audit.jsonl
OFDD_ERROR_LOG_PATH=/var/log/openfdd/error.jsonl
```

# 100.5 Docker edge: json-file caps (not journald for app audit)

Production stacks use **Docker Compose** (`docker/compose.edge.yml`). Container stdout uses the **`json-file` driver with size limits**:

```yaml
logging:
  driver: json-file
  options:
    max-size: "25m"
    max-file: "5"
```

That caps each service at roughly **125 MB** of rotated container logs on disk — the same pattern many teams use instead of unbounded `docker logs`.

**Why audit JSONL stays on disk (not journald)?**

- Append-only files under `workspace/logs/` survive container recreate and bind-mount to the host.
- JSONL lines import cleanly into SIEM without journal field parsing.
- OT incident response often needs **months** of auth/BACnet write history on the edge box.

**When journald still applies:** native systemd units (Caddy, Ollama on bare metal) — use `journalctl -u <unit>` per your Linux runbook. Bridge audit remains in `workspace/logs/` even when uvicorn stdout goes to Docker or `.local-run/`.

# 100.6 Local dev (`run_local.sh`)

| File | Service |
|---|---|
| `workspace/.local-run/bridge.log` | Bridge / uvicorn |
| `workspace/.local-run/commission.log` | BACnet commission agent |
| `workspace/.local-run/caddy.log` | Caddy |
| `workspace/.local-run/fdd_loop.log` | Scheduled FDD |
| `workspace/.local-run/mcp_rag.log` | MCP RAG |
| `workspace/.local-run/ollama.log` | Ollama |

These are **engineering logs**. Structured audit still lands in `workspace/logs/audit.jsonl` when auth is enabled.

## 100.7 Quick checks

```
# Last auth failures
grep 'auth.login' workspace/logs/audit.jsonl | tail -5

# Integrator API (after login)
curl -s -H "Authorization: Bearer $TOKEN" http://127.0.0.1:8765/api/audit/summary | jq .

# Docker edge
docker compose logs bridge --tail 50
```

## 100.8 Comparison to typical web apps

| Practice | Open-FDD |
|---|---|
| Structured auth audit | Yes — JSONL with actor, IP, outcome |
| Login rate limit + lockout audit | Yes |
| Secret redaction in logs | Yes — key-based + agent sanitization |
| Request correlation ID | Yes — on audited mutation paths |
| Centralized stdout JSON | No — stdout is plain text; audit is JSONL |
| Full HTTP access log | No — mutations and auth only (OT noise reduction) |
| Log rotation / size caps | Yes — defaults above + Docker `max-size` |
| SIEM-ready export | Yes — file tail / Filebeat on `workspace/logs/` |

For BACnet write safety and credential generation, see SECURITY.md and Secrets and auth.

## 100.9 Related

- JSON API driver — OpenWeatherMap showcase, env-based API keys (`json_api.env.local`)
- Live site update — `docker compose logs` after upgrades
- Data flow — historian retention (`OFDD_FEATHER_*`) is separate from audit logs

# 101 Fault Codes

# 102 Fault Codes

Open-FDD uses a **check-engine** model: one building-level status (green / yellow / red) with a catalog of fixed fault codes per equipment family.

| Page | Content |
|---|---|
| Convention | Naming, severity, categories |
| AHU faults | Air handling units |
| VAV faults | Terminal units |
| RTU faults | Packaged rooftops |
| Sensor & data quality | Stale, flatline, comms |

Live catalog API: `GET /api/faults/catalog` on the Operator Bridge.

# 103 Operations

# 104 Operations

Runbooks for live edge hosts after the initial Quick Start deploy.

| Page | When to use |
|---|---|
| Live site update | Pull new GHCR tags, preserve `workspace/` |
| Backup and restore | Archive `workspace/` before upgrades |
| Logging and audit | Auth audit trail, rotation, Docker log caps |
| Deployment validation | Post-upgrade smoke and insurance checks |

Site-specific lab notes (example BACnet scope, GL36 rules): Examples & lab notes.

For first-time deploy: Quick Start — Docker.

# 105 LAN hardening

# 106 LAN hardening

Open-FDD targets **trusted building LAN / OT edge** deployment. The Operator Bridge is not intended for direct public internet exposure.

Full mode matrix: Deployment modes.

## 106.1 Auth and bind

| Mode | Bridge bind | Auth |
|---|---|---|
| Local dev | `127.0.0.1` | Optional `OFDD_AUTH_DISABLED=1` on loopback |
| Lab LAN | `0.0.0.0` | Required unless explicit insecure lab flags |
| Edge production | Loopback + **Caddy** on LAN | **Required** — `OFDD_AUTH_SECRET` + role passwords |

Startup **fails** on non-loopback bind without credentials (unless insecure lab flags are set).

## 106.2 Network exposure

- Put **Caddy** on `:80` / `:443`; keep bridge `:8765` on loopback when possible.
- Do not port-forward the Operator Bridge to the public internet without TLS and strong auth.
- Keep BACnet on OT VLANs; management UI on IT VLAN with firewall rules.
- BACnet **writes** are gated by default — BACnet write guard.

## 106.3 Rule Lab

Rule Lab runs **operator-authored Python** in a sandbox. Restrict integrator accounts; review rules before enable on production sites.

## 106.4 TLS and LAN security scans

| Topic | Page |
|---|---|
| Certificates (self-signed / site CA) | TLS and certificates |
| ZAP + Nmap bench smoke | ZAP baseline · `scripts/security/` |
| Ubuntu host + Tenable/Nessus | Linux host hardening · Tenable remediation |
| Release test cycle (maintainers) | Security testing |

## 106.5 Ollama on the edge

Ollama has **no built-in auth**. Bind `OLLAMA_HOST=127.0.0.1:11434` on the host; never expose `:11434` on the OT LAN. Open-FDD bridge reaches Ollama server-side only. Validation: `scripts/security/check_openfdd_exposure.sh`.

# 107 ZAP baseline (local bench)

# 108 ZAP baseline (local bench)

Passive OWASP ZAP baseline against the **pentest production stack** on benserver is the standard LAN security smoke before Acme edge deploys.

Full per-revision workflow (pytest → PR → GHCR → Acme): Developer — security testing cycle.

Packaged scan scripts (Windows + Mac/Linux): scripts/security/README.md.

## 108.1 Run

**On the Open-FDD host:**

```
./scripts/pentest_production_stack.sh start
./scripts/pentest_production_stack.sh verify
```

**From a LAN workstation** (after verify passes):

```
# Windows
.\scripts\security\Run-OpenFddSecurityScan.ps1

# macOS / Linux
./scripts/security/run_openfdd_security_scan.sh --url http://192.168.204.18
```

ZAP target (example): `http://192.168.204.18/` — Caddy on `:80` only; bridge `:8765` stays loopback. TLS bench: see TLS and certificates.

# 108.2 Expected results (HTTP bench mode)

| Finding | Severity | Status |
|---|---|---|
| CSP `style-src 'unsafe-inline'` | Medium | **Accepted** — Vite/ React inline styles; TODO to nonce/ hash |
| Missing COEP | Low | **Accepted** — `require-corp` not enabled (breaks some assets) |
| Port 80 open (Caddy) | Info | **Expected** — intentional LAN entry |
| Ports 8765, 5173, 8000, 8080 closed | — | **Good** |
| Port 443 filtered | — | **Expected** in HTTP mode; use `OFDD_CADDY_MODE=tls` for HTTPS bench |
| Duplicate `Referrer-Policy` / `X-Frame-Options` | Low | **Fixed** — bridge owns headers; Caddy does not duplicate |

# 108.3 Header ownership

| Layer | Sets |
|---|---|
| **Bridge** (`security_headers.py`) | CSP, `X-Frame-Options: DENY`, `frame-ancestors 'none'`, COOP, CORP, Permissions- |

| Layer | Sets |
|---|---|
| | Policy, Referrer-Policy, `X-Content-Type-Options` |
| **Caddy HTTP** | Reverse proxy only; strips `Server` banners |
| **Caddy TLS** | Adds `Strict-Transport-Security` only |

## 108.4 Authenticated scan (later)

Baseline scan hits ~4 public endpoints (`/health`, public agent insight routes). Deep API/dashboard coverage requires ZAP with integrator login — see Authenticated scanning (roadmap).

## 108.5 After fixes

Re-run `pentest_production_stack.sh verify`, then the packaged LAN scan from `scripts/security/`. Confirm response headers show **one** `X-Frame-Options: DENY` and no duplicate `Referrer-Policy`.

# 109 TLS and certificates

# 110 TLS and certificates

Who generates, stores, and renews TLS material for Open-FDD on OT LANs — and how that ties into ZAP/Nmap bench testing.

Developer release cycle: Security testing cycle.

## 110.1 Ownership

| Environment | Who owns certs | Storage |
|---|---|---|
| Local / benserver bench | Developer / maintainer | `workspace/deploy/caddy/certs/` (`cert.pem`, `key.pem`) |
| Ansible edge (Acme, customer sites) | Site integrator | `/etc/openfdd/caddy/` on the VM |

| Environment | Who owns certs | Storage |
|---|---|---|
| | | (playbook-managed or manual) |
| Public internet | **Not in scope** | Open-FDD targets OT LAN; use site PKI or self-signed |

Open-FDD does **not** ship Let’s Encrypt or public CA automation. OT benches use **self-signed** or customer-provided certs.

# 110.2 Generate bench certs (local)

```
./scripts/setup_caddy_certs.sh
# or: ./scripts/setup_caddy_certs.sh --cn openfdd.local --out
workspace/deploy/caddy/certs
```

Then start with TLS mode:

```
OFDD_CADDY_MODE=tls ./scripts/run_local.sh
# or pentest stack with OFDD_CADDY_MODE=tls
```

Caddy serves `:443` and redirects `:80` → HTTPS. Caddy adds **`Strict-Transport-Security`** only; all other security headers remain on the **bridge** (ZAP baseline).

# 110.3 ZAP / Nmap with TLS bench

When `OFDD_CADDY_MODE=tls`:

- Nmap: expect **443 open**, 80 may redirect
- ZAP: use `https://<lan-ip>/` (add `-k` trust or import `cert.pem` on the scan workstation)
- PowerShell: `.\Run-OpenFddSecurityScan.ps1 -TargetUrl "https://192.168.204.18"`
- Bash: `./run_openfdd_security_scan.sh --url https://192.168.204.18`

Self-signed certs produce ZAP TLS warnings — expected on OT LAN.

# 110.4 Renewal

Self-signed certs from `setup_caddy_certs.sh` default to **365 days**. Re-run the script before expiry or bake renewal into site runbooks.

Edge Ansible deploys should document whether the integrator rotates certs manually or via site PKI.

# 110.5 Enterprise CA on Caddy (edge)

When IT requires **clean Nessus SSL plugins** (no self-signed / untrusted chain), install a site-provided certificate on the edge VM instead of `setup_caddy_certs.sh` output.

Typical layout (Ansible / manual):

```
/etc/openfdd/caddy/cert.pem   # full chain (leaf + intermediates)
/etc/openfdd/caddy/key.pem    # private key (0600, root or caddy
user)
```

Ansible TLS mode (`caddy_mode: tls`) already points Caddy at those paths — see `infra/ansible/templates/Caddyfile.j2`.

**Internal enterprise CA workflow:**

1. Generate CSR on the edge host or request cert from site PKI for the operator hostname (e.g. `openfdd.acme.local`).
2. Install `cert.pem` + `key.pem` under `/etc/openfdd/caddy/`.
3. Distribute the **CA root** to admin laptops (trust store) so browsers and ZAP scans validate the chain.
4. Monitor expiry:

```
openssl x509 -in /etc/openfdd/caddy/cert.pem -noout -dates
```

**Caddyfile snippet (TLS, enterprise cert):**

```
:443 {
    tls /etc/openfdd/caddy/cert.pem /etc/openfdd/caddy/key.pem
    header {
        Strict-Transport-Security "max-age=31536000;
includeSubDomains"
    }
    reverse_proxy 127.0.0.1:8765
}
```

Lab/self-signed mode still produces expected **Medium** Tenable findings — document as accepted risk or switch to enterprise CA. See Tenable remediation.

## 110.6 Caddy internal CA mode (lab)

`./scripts/setup_caddy_certs.sh` generates a **self-signed** pair for bench TLS. Trust `cert.pem` on scan workstations (`-k` curl, ZAP import) or accept browser warnings. Not suitable when IT mandates trusted chains.

## 110.7 Secrets

- **Never commit** `key.pem` or production cert bundles
- `workspace/deploy/caddy/certs/` is for local bench only unless your `.gitignore` already excludes generated keys (verify before push)

# 111 Authenticated scanning (roadmap)

# 112 Authenticated scanning (roadmap)

ZAP baseline is **unauthenticated** — it checks public pages, response headers, and a handful of routes reachable without login. That is the right default for every docker image / patch revision smoke on a test LAN.

For **deep** coverage of integrator dashboards, `/api/*` routes, WebSocket sessions, and role-gated writes, plan a separate **authenticated** scan — only on fake/bench stacks with explicit approval.

Release context: Security testing cycle.

## 112.1 Current state (3.0.x)

| Capability | Status |
|---|---|
| ZAP baseline via `scripts/security/` | **Shipped** |
| Pentest creds in `workspace/auth.pentest.local` | **Shipped** (host-local, not in git) |
| ZAP full / active scan automation | **Not shipped** — manual only |

| Capability | Status |
|---|---|
| ZAP authenticated context (login script + session) | **Roadmap** |
| CI running ZAP against PR previews | **Not planned** (LAN bench only) |

## 112.2 Why baseline is not enough

After login, the bridge exposes model editing, BACnet/JSON API configuration, Rule Lab, commissioning export, and write-gated commands. Baseline ZAP never exercises:

- Session cookies / JWT persistence
- CSRF or auth bypass on mutating endpoints
- IDOR on site-scoped resources
- Rate limits and audit logging on failed auth

## 112.3 Planned approach (contributors welcome)

1. **Bench-only target** — same pentest stack; never point active scans at live OT controllers
2. **Login hook** — ZAP `zap-full-scan.py` or Automation Framework with:
   - `POST /api/auth/login` using integrator creds from env (not committed)
   - Regex or JSON extractor for session token / cookie
3. **Scope file** — include `/api/model/*`, `/api/commissioning/*`, dashboard static assets; exclude BACnet UDP and unrelated LAN hosts
4. **Report parity** — extend `scripts/security/` with opt-in `-Authenticated` / `--zap-full` flags and separate report prefix (`22-zap-authenticated-*`)
5. **Gate** — document accepted findings; fail release only on High/Critical regressions

## 112.4 Manual authenticated ZAP today

1. Start pentest stack; note creds in `workspace/auth.pentest.local`
2. Open ZAP desktop or docker with API port exposed
3. Configure **Context** → **Authentication** → script or form-based login to `/api/auth/login`
4. Run **Spider** then **Active Scan** on `http://<lan-ip>/` only
5. Compare with baseline report in `openfdd-security-report/`

Do **not** run active scans against production Acme without a maintenance window and BACnet write lock.

## 112.5 Related

- scripts/security/README.md — baseline setup
- Secrets and auth — env file layout
- LAN hardening — bind and auth modes

# 113 Secrets and auth

# 114 Secrets and auth

## 114.1 Files (gitignored)

| File | Contents |
|---|---|
| `workspace/auth.env.local` | JWT secret, role passwords |
| `workspace/json_api.env.local` | REST API keys for JSON API `${ENV:VAR}` URLs (e.g. OpenWeather `OPENWEATHER_API_KEY`) |
| `workspace/data.env.local` | Historian + **audit log retention** (`OFDD_LOG_*`, `OFDD_FEATHER_*`) |
| `workspace/ollama.env.local` | Ollama URL, model, RAM tier, GPU mode |
| `workspace/mcp.env.local` | MCP RAG sidecar |
| `workspace/caddy.env.local` | Reverse proxy TLS/HTTP mode |
| `workspace/pentest.env.local` | LAN security scan stack |
| `infra/ansible/secrets/<host>.env.local` | SSH deploy secrets (maintainers) |

Copy from `*.example` templates only. Full inventory: Workspace env files.

## 114.2 Required variables (production)

| Variable | Purpose |
|---|---|
| `OFDD_AUTH_SECRET` | HMAC signing (32+ chars) |
| `OFDD_INTEGRATOR_USER` / `PASSWORD` | Rule Lab, bindings |
| `OFDD_OPERATOR_USER` / `PASSWORD` | Operations |

## 114.3 GHCR pulls

Use read-only registry credentials on edge if images are private; rotate tokens periodically.

## 114.4 Backup

Back up `workspace/data/` and auth env in secure operator vault — not in git.

Audit JSONL under `workspace/logs/` may contain client IPs and usernames — treat like security logs. See Logging and audit.

# 115 BACnet write allowlists

# 116 BACnet write allowlists

Supervisory writes require:

1. **Feature flag** enabled in environment (off by default).
2. **Allowlist** file at `workspace/bacnet/write_allowlist.json` (copy from `write_allowlist.example.json`). If the flag is on but the allowlist is missing, writes are **denied**.
3. **Integrator** role on API calls.
4. **Audit** log review after commissioning sessions.

Allowlist entries support `device_instance`, `object_identifier`, `property_identifier`, optional `priority_min`/`priority_max`, `value_min`/`value_max`, and `allowed_values` for binary/multistate targets.

Lab-only override: `OFDD_BACNET_WRITE_ALLOW_ANY=1` (audited; not for production edges).

See operator guide: BACnet write safety.

Never enable blanket write access for agent or integrator automation without human approval per site.

# 117 Linux host hardening

# 118 Linux host hardening

Generic **Ubuntu edge VM** guidance for Open-FDD deployments scanned by IT tools (Tenable/Nessus, Qualys, etc.). This complements application-level LAN hardening and TLS.

Nessus-specific finding tables: Tenable remediation.

## 118.1 Scope

| Layer | Owner | Repo support |
|---|---|---|
| Ubuntu kernel, OpenSSH, apt packages | **Site IT / integrator** | `scripts/security/remediate_ubuntu_host.sh` |
| Docker engine, compose, Caddy | **Integrator** | Ansible + `check_openfdd_exposure.sh` |
| Open-FDD containers (pip/ pyOpenSSL) | **Open-FDD maintainers** | `docker/python-security-constraints.txt`, GHCR images |
| Self-signed TLS on OT LAN | **Expected lab finding** | Enterprise CA path in TLS |

## 118.2 Quick operator workflow

On the edge VM (SSH):

```
        cd ~/open-fdd
        ./scripts/security/check_host_security.sh
        ./scripts/security/check_openfdd_exposure.sh --edge-ip "$
(hostname -I | awk '{print $1}')"
```

Remediate host OS (maintenance window):

```
./scripts/security/remediate_ubuntu_host.sh          # dry-run
sudo ./scripts/security/remediate_ubuntu_host.sh --apply
# if /var/run/reboot-required exists:
sudo reboot
./scripts/security/check_host_security.sh
```

Pull new Open-FDD images after a release that pins container pip/pyOpenSSL:

```
OPENFDD_IMAGE_TAG=latest ./scripts/upgrade_edge_ghcr.sh --limit <inventory_host>
```

## 118.3 Ubuntu version and EOL

Supported targets: **Ubuntu 22.04 LTS** or **24.04 LTS** (64-bit). Rebuild hosts still on **16.04** or **18.04** — `apt upgrade` cannot clear an SEoL finding.

Validate a surprising 16.04 Nessus hit:

```
lsb_release -a
cat /etc/os-release
# Scan credential / stale DNS / container banner mismatch?
```

## 118.4 Kernel and package updates

```
sudo apt update
sudo apt full-upgrade -y
uname -r
test -f /var/run/reboot-required && echo "reboot required"
sudo reboot   # maintenance window
```

Optional status tools:

```
ubuntu-security-status    # or: pro status
apt list --upgradable | grep -i security
```

## 118.5 Host Python (pip, wheel, pyOpenSSL)

Tenable often reports **host** `pip`, `wheel`, and `pyOpenSSL` versions installed by Ubuntu packages or admin `pip install`. Prefer **apt** on the host:

```
python3 -m pip show pip setuptools wheel pyOpenSSL
sudo apt install --only-upgrade python3-pip python3-openssl python3-wheel
```

Open-FDD **containers** pin toolchain versions at image build — see `docker/python-security-constraints.txt`. Rescan after `docker compose pull`.

## 118.6 Ollama — no LAN exposure

Ollama has **no built-in authentication** in typical deployments. Treat LAN exposure as **Critical**.

| Rule | Implementation |
|---|---|
| Bind loopback only | `OLLAMA_HOST=127.0.0.1:11434` in systemd (`ollama.service`) or dev bootstrap |
| Bridge calls server-side | `OFDD_OLLAMA_BASE_URL=http://127.0.0.1:11434` or `host.docker.internal:11434` from containers |
| Never publish `:11434` on `0.0.0.0` | Compose uses `127.0.0.1:11434:11434` when Docker Ollama is enabled |

Validation:

```
        ss -tulpn | grep 11434
        curl -sf http://127.0.0.1:11434/api/tags | head
        curl -sf --max-time 3 http://<edge-lan-ip>:11434/api/tags
# must FAIL from another host
```

Firewall (if management host needs GPU Ollama — rare):

```
        # Example: block LAN → 11434; allow loopback only (adjust
interface names)
        sudo ufw deny 11434/tcp
        sudo ufw allow from 127.0.0.0/8 to any port 11434 proto tcp
```

## 118.7 OpenSSH Terrapin (CVE-2023-48795)

Apply Ubuntu security updates first (`openssh-server`, `openssh-client`). Verify:

```
        ssh -V
        sudo sshd -T | grep -Ei '^(ciphers|macs|kexalgorithms)'
```

Manual algorithm hardening is **optional** and can lock out clients — document rollback (`sshd_config.bak`) before changes. Do not automate cipher stripping in repo scripts.

## 118.8 ICMP timestamp (Low)

Nessus may report ICMP timestamp disclosure. Optional hardening at host firewall or upstream network — low priority for isolated OT VLANs.

Example (nftables — site-specific):

```
        # Block ICMP timestamp / type 13 — verify with IT before applying
        sudo nft add rule inet filter input icmp type timestamp-request
drop
```

## 118.9 TLS and certificate scans

Self-signed or internal CA certificates produce **Medium** Nessus SSL findings (untrusted, self-signed, expiry). That is **expected** for lab/OT encryption-in-transit. For clean SSL plugin results, install a **site enterprise CA** certificate on Caddy — TLS and certificates.

## 118.10 Maintainer security audits (CI parity)

```
        ./scripts/security/run_maintainer_audits.sh
```

Matches GitHub Actions `operator-bridge-security` job where possible.

# 119 Tenable / Nessus remediation

# 120 Tenable / Nessus remediation

Remediation plan for **IT/Tenable.io Nessus** findings on an Open-FDD **OT/LAN edge** deployment (e.g. Acme GL36 lab VM). Open-FDD runs behind a building firewall on Ubuntu with Docker Compose, Caddy, and optional host Ollama.

Related: Linux host hardening, TLS and certificates, LAN hardening, Security testing cycle.

## 120.1 Prioritized action plan

| Priority | Action | Owner |
|---|---|---|
| **P0** | Confirm Ubuntu is **not** 16.04/18.04; rebuild if EOL | IT / integrator |
| **P0** | Bind Ollama to **127.0.0.1** only; verify LAN cannot reach `:11434` | Integrator |
| **P0** | `apt full-upgrade` + **reboot** for kernel USNs (8254, 8278, 8373) | IT / integrator |
| **P1** | Upgrade host `pip`/`wheel`/`pyOpenSSL` via apt or documented pip path | IT / integrator |
| **P1** | Pull GHCR images built with `docker/python-security-constraints.txt` | Integrator |
| **P2** | OpenSSH packages current; review Terrapin after patch | IT |
| **P2** | TLS: accept self-signed findings **or** install enterprise CA cert | IT decision |
| **P3** | ICMP timestamp: optional firewall rule | Network / IT |

## 120.2 Finding classification table

| Nessus finding | Severity | Class | Remediation |
|---|---|---|---|
| Canonical Ubuntu Linux SEoL (16.04.x) | Critical | Host OS / **false positive?** | `lsb_release -a` — rebuild VM on 24.04 LTS if truly 16.04; if host is 22.04/24.04, check scan target IP, credentials, stale asset record |
| | Critical | Deployment config | |

| Nessus finding | Severity | Class | Remediation |
|---|---|---|---|
| Ollama Unauthenticated Access | | | `OLLAMA_HOST=127.0.0.1:11434`; no `0.0.0.0:11434`; firewall deny; see Ollama |
| Ubuntu kernel USN-8254-1 | Critical | Host OS | `sudo apt update && sudo apt full-upgrade -y && sudo reboot` |
| pyOpenSSL 22.0.x < 26.0.0 Buffer Overflow | Critical | Host pip **and** container | Host: `apt upgrade python3-openssl` or pip ≥26; Container: GHCR image with constraints file |
| Ubuntu pip USN-8344-1 | High | Host OS + container build | `apt full-upgrade`; image build upgrades `pip>=26` |
| SSL Certificate Cannot Be Trusted | Medium | **Accepted risk** (lab) / deployment | Self-signed Caddy — use enterprise CA for clean scan |
| SSL Self-Signed Certificate | Medium | **Accepted risk** (lab) | Same as above |
| SSL Certificate Expiry | Medium | Deployment / monitoring | `openssl x509 -dates`; renew `setup_caddy_certs.sh` or site PKI |
| SSH Terrapin (CVE-2023-48795) | Medium | Host OS | Upgrade `openssh-server`; validate `ssh -V` |
| Ubuntu wheel USN-8221-1 | Medium | Host OS | `apt full-upgrade` |
| Ubuntu kernel USN-8278-1, USN-8373-1 | Medium | Host OS | `apt full-upgrade` + reboot |
| pyOpenSSL 0.14.x Security Bypass | Medium | Host + container | Same as pyOpenSSL Critical — upgrade to ≥26 |
| ICMP Timestamp Request | Low | Network / optional host | Firewall drop; often accepted on isolated OT VLAN |

**Class legend:** (1) repo-owned fix · (2) Docker base/image · (3) host OS · (4) deployment/config · (5) accepted risk · (6) validate false positive

# 120.3 Operator commands (edge VM)

## 120.3.1 Validate (read-only)

```
cd ~/open-fdd
./scripts/security/check_host_security.sh
./scripts/security/check_openfdd_exposure.sh --edge-ip "$
```

```
(hostname -I | awk '{print $1}')"
        lsb_release -a
        uname -r
        test -f /var/run/reboot-required && cat /var/run/reboot-
required.pkgs
        python3 -m pip show pip setuptools wheel pyOpenSSL 2>/dev/null |
grep -E '^(Name|Version):'
        ss -tulpn | grep -E ':80|:443|:8765|:11434'
        curl -sf http://127.0.0.1:11434/api/tags | head -3
```

From a **different LAN workstation** (must fail):

```
        curl -sf --max-time 3 http://<edge-ip>:11434/api/tags
```

### 120.3.2 Remediate host (maintenance window)

```
        ./scripts/security/remediate_ubuntu_host.sh              # dry-run
        ./scripts/security/remediate_ubuntu_host.sh --apply
        sudo reboot    # if reboot-required
        ./scripts/security/check_host_security.sh
```

### 120.3.3 Remediate Open-FDD containers

```
        OPENFDD_IMAGE_TAG=latest ./scripts/upgrade_edge_ghcr.sh --limit
<inventory_host>
        docker compose pull && docker compose up -d
```

### 120.3.4 Rescan

After reboot and image pull, request IT **Nessus rescan** of the edge IP. Compare plugin output to the table above.

## 120.4 Ollama unauthenticated access (Critical)

**Not an Open-FDD code bug** — Ollama's API is unauthenticated by design. Risk is **network exposure**.

Repo defaults (3.0.5+):

- Systemd: `Environment=OLLAMA_HOST=127.0.0.1:11434` in `infra/ansible/templates/ollama.service.j2`
- Compose Ollama (optional profile): `127.0.0.1:11434:11434` only
- Acme-style edge: host Ollama + bridge via `host.docker.internal:11434` (container → host loopback path)
- `post_deploy_check.sh` warns if `:11434` listens on `0.0.0.0`

Re-apply Ansible Ollama unit after upgrade:

```
        ansible-playbook -i inventory.yml ollama_bootstrap.yml --limit
<host> -e enable_ollama=true
```

# 120.5 pyOpenSSL and pip in containers

Docker build installs `docker/python-security-constraints.txt` before application requirements:

- `pip>=26.0`, `setuptools>=75.0`, `wheel>=0.45`
- `pyOpenSSL>=26.0.0`, `cryptography>=44.0.0`

CI `pip-audit` runs against bridge + bacnet requirements with the same constraints.

# 120.6 TLS — expected vs production

| Mode | Nessus SSL plugins | When to use |
|---|---|---|
| Caddy HTTP `:80` | No TLS plugins | Dev / HTTP-only VLAN |
| Caddy self-signed `:443` | Medium: untrusted, self-signed, expiry | OT LAN encryption, lab |
| Enterprise CA cert on Caddy | Clean SSL plugins (if chain trusted) | IT mandates clean Nessus |

See TLS — enterprise CA.

# 120.7 Residual findings after remediation

Even with a patched host and loopback-only Ollama, IT may still see:

- **SSL** Medium on self-signed TLS (accepted for lab; document in risk register)
- **ICMP** Low (optional network block)
- **Container** plugins if Nessus scans Docker overlay networks — clarify scan scope with IT

# 120.8 Notes for IT / security team

1. Scan the **edge VM IP**, not individual container IPs, unless container CVE correlation is required.

2. Open-FDD **bridge :8765** should be **loopback-only**; LAN entry is **Caddy :80/:443**.
3. **BACnet UDP 47808** on commission is expected OT traffic — out of scope for web/plugin remediation.
4. Do not expose Ollama **11434** to the building LAN; agent chat is server-side only.
5. Kernel CVEs require **host reboot** — schedule with operations.
6. For authenticated web/API depth, see roadmap Authenticated scanning.

## 120.9 Repo-owned fixes (this document's release)

| Item | Location |
|---|---|
| Container pip/ pyOpenSSL pins | `docker/python-security-constraints.txt`, `docker/Dockerfile` |
| Ollama loopback systemd | `infra/ansible/templates/ollama.service.j2` |
| Host/exposure check scripts | `scripts/security/check_*.sh`, `remediate_ubuntu_host.sh` |
| Post-deploy Ollama bind warn | `infra/ansible/scripts/post_deploy_check.sh` |
| CI pip-audit + constraints | `.github/workflows/ci.yml` |

# 121 Security

# 122 Security

LAN deployment security for the Operator Bridge and BACnet write guard. Open-FDD is for **trusted edge / OT LAN** — not direct public internet exposure.

| Page | Topic |
|---|---|
| LAN hardening | Auth and bind modes |
| ZAP baseline (local bench) | Pentest stack, expected warnings |
| TLS and certificates | Cert ownership, bench TLS mode |
| Linux host hardening | Ubuntu edge VM, SSH, kernel, Ollama bind |
| Tenable / Nessus remediation | IT scan findings, operator runbook |

| Page | Topic |
|---|---|
| Authenticated scanning (roadmap) | Deep ZAP / protected API |
| Secrets and auth | Env files |
| BACnet write allowlists | Supervisory write gates |
| Release & LAN scan cycle | Per-revision ZAP + CI workflow |
| Logging and audit | Auth audit trail, rotation, SIEM |

# 123 API routes

# 124 API routes

REST API served by **`openfdd-bridge`** (default `http://127.0.0.1:8765`). Production: Caddy on `:80` proxies to the bridge.

**OpenAPI:** `GET /docs` and `GET /redoc` when the bridge runs.

**Auth:** JWT via `POST /api/auth/login`. Most routes require `Authorization: Bearer <token>`. Roles: `viewer`, `operator`, `integrator`, `commission`, `agent`, `admin`. BACnet writes require commission role + write safety. Dev: `OFDD_AUTH_DISABLED` on loopback only.

**WebSocket:** `WS /ws/dashboard` — `POST /api/auth/ws-ticket`, then pass the ticket via `Sec-WebSocket-Protocol: ofdd.ws, <ticket>`. Query `?ticket=` is dev-only when `OFDD_WS_ALLOW_QUERY_TICKET=1`. Tickets are short-lived and not interchangeable with the Bearer token.

---

## 124.1 Health & audit

| Method | Path | Role | Description |
|---|---|---|---|
| GET | `/health` | public | Minimal liveness (`ok`, `service`, `version`, `auth_required`) |
| GET | `/health/stack` | auth | Stack traffic-light (verbose URLs/bind with `OFDD_DEBUG_DIAGNOSTICS`) |
| POST | `/api/auth/ws-ticket` | auth | Short-lived WebSocket ticket |
| GET | | auth | Audit counters |

| Method | Path | Role | Description |
|---|---|---|---|
| | `/api/audit/summary` | | |
| GET | `/api/audit/events` | auth | Audit log (query params) |
| GET | `/api/audit/errors` | auth | Recent API errors |

*Exact public set depends on `OFDD_AUTH_DISABLED` and middleware.

## 124.2 Auth

| Method | Path | Description |
|---|---|---|
| POST | `/api/auth/login` | Username/password → JWT |
| GET | `/api/auth/me` | Current user |
| GET | `/api/auth/status` | Auth mode / lockout hints |

## 124.3 Rules & FDD

| Method | Path | Role | Description |
|---|---|---|---|
| GET | `/api/rules/saved` | user | Rule index |
| POST | `/api/rules/save` | operator+ | Create/update rule metadata |
| GET | `/api/rules/saved/{id}/source` | user | Python source |
| PUT | `/api/rules/saved/{id}/source` | operator+ | Write `.py` file |
| GET | `/api/rules/assignments` | user | Point bindings |
| POST | `/api/rules/bind` | integrator+ | Bind rule to point |
| POST | `/api/rules/bindings` | integrator+ | Bulk bindings |
| POST | `/api/rules/batch` | operator+ | Run all enabled rules → `fdd_results.json` |

| Method | Path | Role | Description |
|---|---|---|---|
| GET/POST | `/api/rules/drafts` | user | Draft rules |

## 124.4 Rule Lab playground

| Method | Path | Description |
|---|---|---|
| POST | `/api/playground/lint` | AST lint Python rule |
| POST | `/api/playground/test-rule` | Run `apply_faults_arrow()` on feather PyArrow window |
| POST | `/api/playground/run-script` | Execute script mode |
| GET | `/api/playground/sample-frame` | Sample DataFrame for editor |

## 124.5 BRICK / data model

| Method | Path | Description |
|---|---|---|
| GET | `/api/model/sites` | Site list |
| GET | `/api/model/tree` | Equipment tree (SPARQL) |
| GET | `/api/model/graph` | RDF graph fragment |
| GET | `/api/model/scope` | Scoped query |
| GET | `/api/model/export` | Export TTL/JSON |
| GET | `/api/model/health` | Model point/equipment counts |
| GET | `/api/model/ttl` | TTL metadata |
| POST | `/api/model/sync-ttl` | Regenerate TTL from JSON |
| POST | `/api/model/import` | Import model payload |
| POST | `/api/model/sites` | Create site |
| GET/POST | `/api/model/bacnet-sync` | BACnet ↔ model sync state |

## 124.6 BACnet

| Method | Path | Role | Description |
|---|---|---|---|
| GET | `/config/bacnet` | read | Commission URL / config |
| GET | `/api/bacnet/commission/status` | read | Commission agent health |
| GET | `/api/bacnet/server/points` | read | Server point table |
| GET | `/api/bacnet/inventory` | read | Discovered inventory |
| POST | `/api/bacnet/whois` | commission | Who-Is |
| POST | `/api/bacnet/discover` | commission | Discover devices |
| POST | `/api/bacnet/read` | commission | Read property |
| POST | `/api/bacnet/read-multiple` | commission | RPM read |
| POST | `/api/bacnet/write` | write | Supervisory write (gated) |
| POST | `/api/bacnet/import-to-model` | integrator | Inventory → BRICK |
| POST | `/api/bacnet/driver/sync-discovery` | commission | Sync driver tree |
| POST | `/api/bacnet/poll/once` | commission | Trigger poll cycle |
| GET | `/api/bacnet/poll/status` | read | Poll worker status |

| Method | Path | Role | Description |
|---|---|---|---|
| POST | `/ingest/bacnet` | read | Ingest samples into feather |

Bind address: `BACNET_BIND` in commission env — see BACnet network setup. Operator guide: Discover and read.

PATCH | `/api/bacnet/driver/point` | commission | Enable poll + interval on point |
PATCH | `/api/bacnet/driver/device` | commission | Enable poll for all points on device |
PATCH | `/api/bacnet/driver/device/remap` | commission | Change instance and/or address |
DELETE | `/api/bacnet/driver/point/{point_id}` | commission | Remove point from registry |
DELETE | `/api/bacnet/driver/device/{device_instance}` | commission | Remove device |
DELETE | `/api/bacnet/driver/registry` | commission | Clear CSVs + model sync |

## 124.7 Faults (check-engine)

| Method | Path | Description |
|---|---|---|
| GET | `/api/faults/catalog` | Fixed fault codes |
| GET | `/api/faults/status` | GREEN/YELLOW/RED aggregate |
| GET | `/api/faults/tree` | Fault tree by equipment |
| GET | `/api/faults/code/{code}` | Code metadata |

## 124.8 Timeseries & sites

| Method | Path | Description |
|---|---|---|
| GET | `/api/timeseries/sites` | Feather sites |
| GET | `/api/timeseries/series` | Series metadata |
| GET | `/api/timeseries/plot` | Plot payload |
| GET | `/api/timeseries/readings` | Raw readings |
| GET | `/api/sites` | Site list |
| GET | `/api/sites/{site_id}/frame` | Wide frame for site |

## 124.9 Building & host

| Method | Path | Description |
|---|---|---|
| GET | `/api/building/status` | Building summary |
| GET | `/api/building/alerts` | Active alerts |
| GET | `/api/host/stats` | CPU/mem/disk |

---

## 124.10 Local agent (Ollama)

Prefix **`/openfdd-agent`**. Role **`agent`** for tool execution.

| Method | Path | Description |
|---|---|---|
| GET | `/openfdd-agent/context` | Composed prompt context |
| GET | `/openfdd-agent/tools` | Tool manifest |
| POST | `/openfdd-agent/tool` | Execute tool (e.g. `rules.save`) |
| GET | `/openfdd-agent/ollama/health` | Ollama reachability |
| GET | `/openfdd-agent/building-insight` | Check-engine narrative context |
| GET | `/openfdd-agent/zone-temps` | Zone temperature insight |
| POST | `/openfdd-agent/chat` | Chat completion (catalog-bound) |

Optional host Ollama for building insight — not required for core FDD. Configure via compose profile or host service; see Architecture overview.

---

## 124.11 Modbus

| Method | Path | Role | Description |
|---|---|---|---|
| GET | `/api/modbus/driver/tree` | operator+ | Commissioning tree |

| Method | Path | Role | Description |
|---|---|---|---|
| GET | `/api/modbus/poll/status` | operator+ | Poll worker status |
| POST | `/api/modbus/poll/once` | integrator+ | Force poll cycle |
| POST | `/api/modbus/read_registers` | operator+ | One-shot register read |
| POST | `/api/modbus/read_and_store` | operator+ | Read + CSV + feather (`source=modbus`) |
| POST | `/api/modbus/refresh` | operator+ | Re-read one register |
| PATCH | `/api/modbus/register/poll` | integrator+ | Enable poll interval |

See Driver framework.

## 124.12 JSON API (HTTP/HTTPS)

| Method | Path | Role | Description |
|---|---|---|---|
| GET | `/api/json-api/env/status` | operator+ | Env var configured flags (no secret values) |
| POST | `/api/json-api/presets/openweather` | integrator+ | Register 3-point OpenWeather bundle + optional poll |
| GET | `/api/json-api/driver/tree` | operator+ | Endpoints grouped by host |
| GET | `/api/json-api/poll/status` | operator+ | Poll worker status |
| POST | `/api/json-api/poll/once` | integrator+ | Force poll cycle |
| POST | `/api/json-api/request` | operator+ | One-shot GET/POST (no store) |
| POST | `/api/json-api/read_and_store` | operator+ | Request + CSV + feather (`source=json_api`) |
| POST | `/api/json-api/refresh` | operator+ | Re-fetch endpoint |
| PATCH | `/api/json-api/endpoint/poll` | integrator+ | Enable poll interval |
| DELETE | | integrator+ | Remove endpoint |

| Method | Path | Role | Description |
|---|---|---|---|
| | `/api/json-api/endpoint/{point_id}` | | |

**Outbound auth:** `auth_type` = `none` | `bearer` | `basic`; optional `verify_tls` (default true). See JSON API driver.

---

## 124.13 PyPI package (separate)

Library-only HTTP is **not** bundled. See Python package for `open_fdd.arrow_runtime` (Arrow rules offline).

---

## 124.14 Errors

JSON error bodies; stack traces hidden unless `OFDD_DEBUG_TRACEBACKS=1`. See LAN hardening.

# 125 Python package

# 126 Python package (`open-fdd`)

Install from PyPI:

```
        pip install open-fdd
# Arrow runtime + playground (default 3.0.1+)
        pip install "open-fdd[ml]"
# optional numpy/sklearn for offline graph ML experiments
```

## 126.1 Modules

| Module | Purpose |
|---|---|
| `open_fdd.arrow_runtime` | **Default** — `apply_faults_arrow` on PyArrow Tables, cookbook masks, column maps |
| `open_fdd.playground` | Rule Lab lint/compile helpers for Arrow rules |

Retired in 3.0.1+: `open_fdd.engine` (YAML/pandas `RuleRunner`) is **not** shipped on PyPI. Use Operator Bridge Rule Lab or import rules from `workspace/data/rules_py/`.

## 126.2 When to use package-only

| Scenario | Use |
|---|---|
| Lint/test Arrow rules offline | `open_fdd.arrow_runtime` + `open_fdd.playground` |
| Graph ML experiments (Layer B) | `open-fdd[ml]` in `experiments/` (see issue #211) |
| Full building operator stack | **Docker** Operator Bridge |

## 126.3 Versioning

Publish: git tag `open-fdd-vX.Y.Z` → GitHub Actions **Publish open-fdd**.

```
import open_fdd
print(open_fdd.__version__)
```

## 126.4 Offline Arrow example

```
import pyarrow as pa
import pyarrow.compute as pc
from open_fdd.arrow_runtime import run_arrow_rule

code = '''
import pyarrow.compute as pc

def apply_faults_arrow(table, cfg, context=None):
    return pc.greater(table["SAT"], float(cfg["high"]))
'''
table = pa.table({"SAT": [70.0, 90.0, 88.0]})
result = run_arrow_rule(code, table, {"high": 85})
print(result.flagged_count)
```

API surface: `open_fdd/arrow_runtime` docstrings and Arrow recipes.

# 127 Configuration reference

# 128 Configuration reference

## 128.1 Bridge / auth

| Variable | Default | Notes |
|---|---|---|
| `OFDD_BRIDGE_HOST` | `127.0.0.1` | `0.0.0.0` for LAN |
| `OFDD_AUTH_SECRET` | — | Required for LAN |
| `OFDD_AUTH_DISABLED` | `0` | Localhost dev only |
| `OFDD_INSECURE_LAN_DEV` | `0` | Lab only with auth disabled |

## 128.2 BACnet

| Variable | Notes |
|---|---|
| `BACNET_BIND` | `ip/mask:47808` in `commission.env` |
| `OFDD_BACNET_WRITE_ENABLED` | Default off — site-specific name may vary |

## 128.3 Docker edge

| Variable | Notes |
|---|---|
| `OPENFDD_IMAGE_TAG` | GHCR tag for Ansible deploy |
| `openfdd_docker_pull_from_ghcr` | Ansible host_var |

Rule Lab Python rules use `rules_store.json` + `workspace/data/rules_py/*.py`. Legacy YAML schema (`docs/config_schema.json`) is archived — not used by Operator Bridge 3.0.1+.

# 129 Glossary

# 130 Glossary

| Term | Definition |
|---|---|
| **Operator Bridge** | FastAPI service serving API + static dashboard |
| **Rule Lab** | UI and APIs for Arrow `apply_faults_arrow()` rules |
| **Feather store** | On-disk columnar historian under `workspace/data/feather_store/` |
| **fdd_input** | Brick tag linking a point to a rule binding role |
| **Check-engine** | Building-level GREEN/YELLOW/RED fault summary |
| **Commission** | BACnet discovery/read (and optional write) agent container |
| **Poll driver** | Scheduled BACnet RPM reads into historian |
| **GHCR** | GitHub Container Registry — `ghcr.io/bbartling/openfdd-*` |

# 131 Workspace env files

# 132 Workspace env files

All live under `workspace/`. Copy from `*.example` templates; committed `*.local` files are gitignored.

| File | Loaded by | Purpose |
|---|---|---|
| `auth.env.local` | `run_local.sh`, Docker compose, Ansible | JWT secret (`OFDD_AUTH_SECRET`), integrator/operator/agent passwords, `OFDD_AUTH_DISABLED` for localhost dev only |
| `data.env.local` | `run_local.sh`, bridge, feather maintain | Feather historian caps (`OFDD_FEATHER_*`), audit/error log rotation |

| File | Loaded by | Purpose |
|---|---|---|
| | | (`OFDD_LOG_*`), BACnet discovery mutation toggle |
| `json_api.env.local` | bridge, Ansible edge sync | API keys for JSON API driver `${ENV:VAR}` substitution (e.g. `OPENWEATHER_API_KEY`) |
| `ollama.env.local` | `run_local.sh`, bridge | Ollama base URL, model name, RAM tier, GPU mode, timeouts — see Ollama and analytics |
| `mcp.env.local` | `run_local.sh`, MCP sidecar | Enable MCP RAG REST (`OFDD_MCP_*`), index path |
| `caddy.env.local` | `run_local.sh`, Ansible edge | Reverse proxy mode (`http` / `tls` / `off`), TLS CN, port overrides |
| `pentest.env.local` | `pentest_production_stack.sh` | Production-like LAN scan stack: CORS, auth, passive BACnet defaults |

## 132.1 Ansible / edge extras

| Path | Purpose |
|---|---|
| `infra/ansible/secrets/<host>.env.local` | SSH deploy credentials, GHCR pull tokens, site-specific BACnet bind — maintainers only |

## 132.2 Related docs

- Secrets and auth — required production variables
- Logging and audit — `OFDD_LOG_*` detail
- JSON API driver — `${ENV:VAR}` pattern
- Configuration reference — bridge runtime flags not in env files

# 133 Appendix

# 134 Appendix

Reference material for integrators and library users.

| Page | Content |
|---|---|
| API routes | Operator Bridge REST/WebSocket |
| Python package | PyPI `open-fdd` |
| Configuration reference | Key env variables |
| Workspace env files | `workspace/*.env.local` inventory |
| Glossary | Terms |

# 135 GL36 lab site (Acme)

# 136 GL36 lab site (Acme)

> **Lab note only.** This documents a maintainer **example edge host** (site `acme`, building `vm-bbartling`) used for Guideline 36 supervisory FDD development. Replace inventory host names, IPs, and BACnet ranges for your site.

Reference for **ASHRAE Guideline 36** supervisory monitoring on a live lab edge. Rule code lives in `workspace/data/rules_py/`; commissioning uses **Arrow constants** (no `config.json`).

Trim & respond Niagara reference: README_TRIM_RESPOND.

## 136.1 BACnet poll scope

Device filter (`infra/ansible/scripts/acme_trim_devices.sh`):

| Range | Equipment |
|---|---|
| 1–100 | JCI VMA/VAV |
| 1100 | RTU-01 (AHU) |
| 1002 | Hot water plant |
| 11000–13000 | Trane UC210 VAV |

Poll interval: **60 s**. Commission from control machine:

```
cd infra/ansible
POLL_INTERVAL=60 ./scripts/acme_commission_gl36.sh
```

Points land in `edge_backup/local/acme/vm-bbartling/points.csv` for Ansible sync.

# 136.2 Example FDD rules (Arrow)

## 136.2.1 Duct static pressure high (AHU)

```
        """AHU duct static high — GL36 trim advisory."""

        import pyarrow.compute as pc

        VALUE_COLUMN = "duct-static"
        HIGH_INWC = 1.20
        LOOKBACK_HOURS = 3


        def _kit_lookback_stats(table, *, hours=None):
            h = hours if hours is not None else LOOKBACK_HOURS
            ts = pc.cast(table["timestamp"], "timestamp[us, UTC]")
            tmin, tmax = pc.min(ts).as_py(), pc.max(ts).as_py()
            span_h = (tmax - tmin).total_seconds() / 3600.0 if tmin and
tmax else 0.0
            print(f"lookback={h}h rows={table.num_rows} start={tmin}
stop={tmax} span={span_h:.2f}h")


        def apply_faults_arrow(table, cfg, context=None):
            _kit_lookback_stats(table)
            return pc.greater(pc.cast(table[VALUE_COLUMN], "float64"),
HIGH_INWC)
```

## 136.2.2 SAT too cold vs setpoint

```
        """AHU SAT too cold vs setpoint."""

        import pyarrow.compute as pc

        SAT_COLUMN = "sa-t"
        SP_COLUMN = "sa-t-sp"
        COLD_DELTA_F = 5.0


        def apply_faults_arrow(table, cfg, context=None):
            sat = pc.cast(table[SAT_COLUMN], "float64")
            sp = pc.cast(table[SP_COLUMN], "float64")
            return pc.less(pc.subtract(sat, sp), -COLD_DELTA_F)
```

## 136.2.3 Site rule stubs in the repo

| Rule file | Use |
|---|---|
| `bench_stat_zn-t_flatline_1h.py` | Zone temp flatline — |

| Rule file | Use |
|---|---|
| | bind `ZN-T` per VAV |
| `acme_duct_static_flatline_1h_gl36_duct_t_r_input.py` | RTU duct static flatline |
| `oat_vs_web_spread_1h.py` | Local OAT vs OpenWeather JSON API |

### 136.2.4 Local OA-T vs web weather (JSON API)

1. Control machine: `workspace/json_api.env.local` with `OPENWEATHER_API_KEY`
2. JSON API tab → **OpenWeatherMap** preset → **Register** (default poll **30 min**)
3. Bind BACnet local OAT in the model (`oa-t` historian column)
4. Enable `oat_vs_web_spread_1h.py` in Rule Lab — flags when `|oa-t − web-oat-t| > 8 °F`

## 136.3 Deploy / upgrade (Ansible, this lab only)

```
        cd infra/ansible
        OPENFDD_IMAGE_TAG=latest ./deploy.sh docker --limit
<inventory_host> -e enable_bacnet_poll_driver=true
```

Full UI + image upgrade from control machine:

```
        OPENFDD_IMAGE_TAG=latest ./scripts/upgrade_edge_full.sh --limit
<inventory_host>
        infra/ansible/scripts/post_deploy_check.sh --limit
<inventory_host> --full
```

Do **not** rsync dev `auth.env.local` to the edge (`deploy_sync_auth: false` in private host_vars). Regenerate credentials on the edge host if login fails after upgrade.

## 136.4 AI-assisted data modeling (lab workflow)

1. BACnet poll running (~340 GL36 points @ 60 s on this lab host).
2. **Model & assignments** → export commissioning JSON for LLM-assisted binding.
3. Import returned JSON via commissioning import.

4. Pin FDD rules on `ZN-T`, `DA-T`, `duct-static`, `OAT`, etc. — export shows readable names in `fdd_rules_linked`.

Model seed: `workspace/data/acme_gl36_model.json`.

# 137 Examples & lab notes

# 138 Examples & lab notes

Site-specific commissioning notes, GL36 examples, and maintainer lab workflows. These pages are **not** required for a generic edge deployment.

| Page | Description |
|---|---|
| GL36 lab site (Acme) | Example real-building poll scope, Arrow rule stubs, Ansible deploy commands |

For production IT operators, start with Quick Start and Operations.

# 139 Central collection

# 140 Portfolio central collection

Edge Open-FDD sites emit **interval summaries** (not raw historian by default). Central portfolio ingests, stores, and aggregates for Dash and the AI tuning agent.

## 140.1 Interval summary schema

Defined in `portfolio/store/interval_schema.py` as `FaultIntervalSummary`:

- Identity: `site_id`, `building_id`, `equipment_id`, `equipment_family`
- Window: `timestamp_start`, `timestamp_end`
- Fault: `fault_code`, `canonical_id`, `severity`, `confidence`
- Activity: `active_minutes`, `percent_active`, `occurrence_count`
- Evidence: `evidence_json` (summarized, not full trend)

- Tuning: `tuning_version`, `tuning_params_hash`
- Operator: `operator_status` (`new` | `acknowledged` | `resolved` | `false_positive` | …)

## 140.2 Edge export formats

- JSONL (append-only events)
- Parquet / Feather (batch intervals)

## 140.3 Collect today (rollup API)

```
source infra/ansible/secrets/acme.env.local
python3 scripts/portfolio_collect.py
```

## 140.4 Tuning proposals

Human-reviewable proposals use `TuningProposal` in the same module. The AI agent **never** auto-writes BACnet points — see `docs/portfolio/ai-agent-tuning.md` (3.0.19).

# 141 Arrow-native runtime

# 142 Arrow-native runtime

Open-FDD 3.0 executes Rule Lab rules on **PyArrow tables** end to end.

```
feather_store.read_site_table()
  → fdd_runner / playground.run_arrow_table()
  → apply_faults_arrow(table, cfg, context)
  → pyarrow.compute masks
```

## 142.1 Authoring contract

```
import pyarrow.compute as pc

def apply_faults_arrow(table, cfg, context=None):
    return pc.greater(table["zone_temp"], cfg["max_zone_temp"])
```

Cookbook helpers: `open_fdd.arrow_runtime.cookbook` (flatline, spread, OOB, after-hours fan).

Script-mode analytics rules receive `table` (PyArrow) and `cfg` in the sandbox — not pandas DataFrames.

## 142.2 Package layout

| Module | Role |
|---|---|
| `open_fdd.arrow_runtime.backend` | Execute rule code, batch chunks |
| `open_fdd.arrow_runtime.cookbook` | Shared fault masks |
| `open_fdd.arrow_runtime.windows` | Rolling min/max, consecutive-true |
| `open_fdd.playground.arrow_templates` | Rule Lab starter templates |